

Implementation of the Dynamic Time Warping Algorithm for Bioacustic Detection

by

Elisabeth Christina Bosch

Dissertation presented for the degree of Electrical and Electronic Engineering in the Faculty of Engineering at Stellenbosch University

Supervisor: Prof. Jaco Versfeld

2026

Acknowledgements

* Supervisor X.
* Funding Y.

> *A very long quote*
> *from someone very important.*
> – Mr M

Thank you to Leanne for helping to prepare this template.

Declaration

By submitting this dissertation electronically, I declare that the entirety of the work contained therein is my own, original work, that I am the sole author thereof (save to the extent explicitly otherwise stated), that reproduction and publication thereof by Stellenbosch University will not infringe any third party rights and that I have not previously in its entirety or in part submitted it for obtaining any qualification.

Date: day October 2026

Copyright © 2026 Stellenbosch University
All rights reserved.

Abstract

This project investigated the use of dynamic time warping (DTW) as a detection system for Bryde's whale calls in False Bay. Three feature extraction methods were considered and compared: the short-time Fourier transform (STFT), mel-frequency cepstral coefficients (MFCCs), with and without first and second derivatives, and linear predictive coding coefficients (LPCCs). Each feature extraction method was tested using both 5 templates per call type and 10 templates per call type.

During hyperparameter tuning, the per-call type thresholds, feature frame length and vote fraction for each model were selected. Selection decisions were made based on the highest f1-scores for each tested setting. After tuning, call detection was performed on a full 72-hour audio file. The performance of each model was analysed using precision, recall, f1-score and average precision (AP).

The STFT model outperformed the other two models with a maximum f1-score of 0.792 and a maximum AP of 0.860. Precision was consistently lower than recall across all models, indicating that the models were more lenient during evaluation than intended in hyperparameter tuning.

Despite limitations caused by data scarcity and labelling inconsistencies, the DTW detection system was able to efficiently detect calls. This was especially true for the STFT feature extraction method.

Table of Contents

Abstract iii

Table of contents iv

List of figures vii

List of tables viii

Nomenclature x

Acronyms and abbreviations . . . x

Definitions . . . xi

1. Introduction 1

1.1. Background . . . 1

1.2. Problem Statement . . . 2

1.3. Objectives . . . 2

1.4. Scope . . . 2

1.5. Layout of report . . . 3

2. Literature 4

2.1. Bryde's Whales . . . 4

2.2. Dynamic Time Warping . . . 5

2.3. Feature Extraction Methods . . . 8

2.3.1. Short-Time Fourier Transform . . . 8

2.3.2. Mel-Frequency Cepstral Coefficients . . . 9

2.3.3. Linear Predictive Coding Coefficients . . . 10

2.4. Similar Studies . . . 11

2.4.1. Classification of Killer Whale Vocalisations using DTW . . . 11

2.4.2. Analysis of Template-Based Detection Algorithms for Bryde's Whales 12

2.4.3. Detection of real-time fin and blue whale calls . . . 12

2.4.4. Hidden Markov Model for Bryde's Whale short pulse calls . . . 13

2.4.5. Speech Recognition using DTW . . . 13

2.5. Chapter Summary 13

3. Dataset **15**
3.1. Background 15
3.2. Labelling 15
3.3. Data Splitting 16
3.3.1. Template Selection 16
3.3.2. Hyperparameter Tuning Template Selection 16
3.3.3. Evaluation Data Selection 17
3.4. Chapter Summary 17

4. System Design **18**
4.1. Overview 18
4.2. Feature Extraction Methods 18
4.2.1. Short-Time Fourier Transform 18
4.2.2. Mel-Frequency Cepstral Coefficients 19
4.2.3. Linear Predictive Coding Coefficients 19
4.3. DTW Methodology 20
4.4. Hyperparameter Tuning 21
4.5. Evaluation Procedure 24
4.5.1. Detection Methodology 24
4.5.2. Matching Methodology 27
4.6. Evaluation Metrics 27
4.7. Chapter Summary 28

5. Results **29**
5.1. Experiment 1: STFT Features 29
5.2. Experiment 2: MFCC Features 31
5.3. Experiment 3: LPCC Features 36
5.4. General Observations 38
5.5. Chapter Summary 40

6. Conclusion **41**
6.1. Summary 41
6.2. Objectives 41
6.3. Limitations 41
6.4. Conclusion 42
6.5. Future Work 42

References **43**

A. Project Planning Schedule **48**

B. Graduate Attributes **49**

B.1. Complex Engineering Problem 49
B.2. GA 1: Problem Solving 50
B.3. GA 2: Application of Scientific and Engineering Knowledge 50
B.4. GA 3: Engineering Design 50
B.5. GA 4: Investigations, experiments and data analysis 50
B.6. GA 5: Use of Engineering Tools 51
B.7. GA 6: Professional and Technical Communication 51
B.8. GA 8: Individual Work 51
B.9. GA 9: Independent Learning Ability 51
B.10. Sustainable Development 51
B.11. Critical Thinking 52
B.12. Emerging Technologies 52

C. Use of Artificial Intelligence (AI) Tools **53**

C.1. AI Work Declaration 53
C.2. Own Work Declaration 53

List of Figures

2.1. Examples of how each call type looks: single-tone, multi-tone, burst-tonal and tonal downsweep. The spectrograms for each call type were printed based on labelled calls from the dataset [1]. 4
2.2. An example of the optimal warping path between two vector features, X with length N = 4 and Y with length M = 6 5
2.3. An example for the three possible options for min as well as an example of the optimal warp path found for signals X and Y assuming X=[0, 2, 1, 0] and Y=[0, 0, 0.5, 2, 1, 0] 7
2.4. Example of a subsequence DTW alignment between two signals $X = [x_1, x_2, x_3, x_4]$ and $Y = [y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9, y_{10}]$ 7
2.5. The mel-scale filter bank is made up of 20 to 40 triangular filters arranged according to the mel scale. 9

4.1. Software flow diagram describing the hyperparameter selection procedure. . . 21
4.2. Software flow diagram describing the evaluation procedure. 24
4.3. Illustration of per call type matrices containing the bottom row of the cost matrix for each template in the given call type for k = 1. The green, purple and pink blocks are the k_{th} lowest cost per frame for each matrix. These correspond to Figure 4.4 The blue block is the detected end frame as it appears in the template analysed to find the start frame. This corresponds to Figure 4.5. 25
4.4. Illustration of the k_{th} best costs matrix. The coloured blocks correspond to those in Figure4.3 26
4.5. Illustration of the cost arrays used to find the start and end frames without normalisation. The coloured blocks corresponds to those in Figure 4.3 and 4.4. 26

5.1. The PR curves for STFT 31
5.2. The PR curves for MFCC 35
5.3. The PR curves for MFCC with the first derivative added 35
5.4. The PR curves for MFCC with the first and second derivatives added 35
5.5. The PR curves for LPCC 37

5.6. Examples of random samples of FP and TP for which some FP could be unlabelled calls for the 10-template experiment for the STFT model. 39
5.7. This shows for each model where FP calls were detected in relation to labelled calls over the entire 72-hour audio file. 39

List of Tables

3.1. Dataset counts by date and category 16

5.1. Calibration results for the STFT model: f1-scores and thresholds per call type, average f1-score and selected frame length (in seconds) for 5 and 10 templates per call type. 29

5.2. Vote fraction selection results for the STFT model: precision, recall and f1-score at the chosen vote fraction for 5 and 10 templates per call type. . . . 30

5.3. Final evaluation results for the STFT model: precision, recall, f1-score and average precision (AP) for 5 and 10 templates per call type. 30

5.4. Calibration results for the MFCC model: f1-scores and thresholds per call type, average f1-score and selected frame length (in seconds) for each feature set (no derivatives, first derivatives and first and second derivatives) with 5 and 10 templates per call type. 32

5.5. Vote fraction selection results for the MFCC model: precision, recall and f1-score at the chosen vote fraction for each feature set with 5 and 10 templates per call type. 33

5.6. Final evaluation results for the MFCC model: precision, recall, f1-score and average precision (AP) for each feature set with 5 and 10 templates per call type. 34

5.7. Calibration results for the LPCC model: f1-scores and thresholds per call type, average f1-score and selected frame length (in seconds) for 5 and 10 templates per call type. 36

5.8. Vote fraction selection results for the LPCC model: precision, recall and f1-score at the chosen vote fraction for 5 and 10 templates per call type. . . . 36

5.9. Final evaluation results for the LPCC model: precision, recall, f1-score and average precision (AP) for 5 and 10 templates per call type. 37

5.10. Summary of the highest f1-score and AP achieved by each feature extraction method, with the corresponding number of templates per call type. 38

A.1. Project Schedule 48

Nomenclature

Acronyms and Abbreviations

AI	Artificial Intelligence
AP	
DCT	Discrete Cosine Transform
DTW	Dynamic Time Warping
FFT	Fast Fourier Transform
FN	False Negative
FP	False Positive
FT	Fourier Transform
HMM	Hidden Markov Model
LPC	Linear Predictive Coding
LPCC	Linear Predictive Coding Coefficients
MFCC	Mel-Frequency Cepstral Coefficients
PR	Precision-Recall
ROC	Receiver Operating Characteristic
SD	Secure Digital
SNR	Signal-to-Noise Ratio
STFT	Short-Time Fourier Transform
TN	True Negative
TP	True Positive
VQ	Vector Quantisation
WAV	Waveform Audio File Format

Definitions

C

class imbalance A situation in which one class occurs far more often than another. In this project, segments of background noise greatly outnumber whale calls

D

downsampling Reducing the sampling frequency of a signal by first removing the frequencies that the lower rate cannot represent and then discarding samples. Here, the audio was downsampled from 8 kHz to 1 kHz

F

f1-score A single measure of detection performance, calculated as the harmonic mean of precision and recall

frame duration The length, in seconds, of the window of audio from which one feature vector is calculated. Also referred to as the frame length

H

Hamming window A window function that tapers the edges of each frame of audio before the Fourier transform is calculated, which reduces spectral leakage

hop length The distance, in samples or seconds, between the starts of consecutive frames. It equals the frame length minus the window overlap

hydrophone An underwater microphone used to record sound in water

hyperparameter A setting of the detection system that is chosen before detection is performed, such as the frame length, call type thresholds or vote fraction, rather than calculated from the audio itself

N

Nyquist frequency The highest frequency that can be represented without aliasing at a given sampling frequency, equal to half the sampling frequency

P

precision The proportion of detections that correspond to labelled calls, calculated as the number of true positives divided by the sum of the true positives and false positives

R

RECALL The proportion of labelled calls that are detected, calculated as the number of true positives divided by the sum of the true positives and false negatives

S

SAMPLING FREQUENCY The number of samples of a signal taken per second, measured in hertz. Also referred to as the sampling rate

SPECTRAL LEAKAGE The spreading of energy from one frequency into neighbouring frequencies, which occurs because the Fourier transform is calculated over a segment of finite length

SPECTROGRAM A visual representation of how the frequency content of a signal changes over time, with time on the horizontal axis, frequency on the vertical axis and magnitude shown by colour

T

TEMPLATE A short labelled example of a call, or of background noise, that is compared with segments of audio using dynamic time warping

V

VOTE FRACTION The fraction of the templates for a call type that must register a detection at a point in the audio before that point is classified as a call

Chapter 1

Introduction

1.1 Background

Van Waerebeek et al. [2] reported indications since the late 1990s that whales have been struck by shipping vessels. Today this has become a larger concern, as vessel speeds and traffic have substantially increased. Bryde's whales are found to migrate in smaller regions instead of the usual large-scale migrations of other whale species. Faustmann et al. [3] report that such a whale colony in False Bay, South Africa has also been placed in possible danger due to fishing activities in their migration region.

Detection and classification systems for whale calls can be used to monitor the migration patterns of specific whale colonies throughout the year. The information gained from such analysis can help determine when vessel traffic should be diverted for the safety of a whale population. These types of systems have been implemented on different whale colonies, as discussed by Betz et al. [4], but the migration patterns of the False Bay colony have not yet been thoroughly documented. Due to the variations in call types between different colonies of some whale species, such as killer whales (Deecke et al. [5]) and blue whales (McDonald et al. [6]), models trained for other Bryde's whale colonies would not be completely trustworthy for the detection of calls within the False Bay Bryde's whale colony, as the universality of calls cannot be assumed. Therefore, a model trained specifically for the False Bay colony would be beneficial.

Brown et al. [7] showed that dynamic time warping (DTW) is an effective method for classifying killer whale vocalisations. Given the limited labelled data for the False Bay colony, DTW is considered as Herreilers et al. [8] found that it requires less input data than other detection algorithms.

1.2 Problem Statement

This project will investigate whether dynamic time warping (DTW) applied to two-dimensional time-frequency representations can effectively detect Bryde's whale calls in False Bay, as measured by precision, recall and the f1-score. Three feature extraction methods will be compared in terms of detection performance. These will be the short-time Fourier transform (STFT), mel-frequency cepstral coefficients (MFCC) and linear predictive coding coefficients (LPCC).

1.3 Objectives

1. Implement a DTW-based detection system using three feature extraction methods: the short-time Fourier transform (STFT), mel-frequency cepstral coefficients (MFCCs) and linear predictive coding coefficients (LPCC).

2. Tune the feature frame length, call type thresholds and vote fraction for each method based on f1-score.

3. Evaluate the detection performance of each configuration using precision, recall, f1-score and average precision (AP).

1.4 Scope

Elements not considered within this project include the classification of call types after detection, the detection of tonal downsweep calls, detection algorithms other than DTW, feature extraction methods other than STFT, MFCC and LPCC, the accuracy of detected start and end times, the detection of frequencies above 500 Hz.

Call type classification was not considered due to discontinuities in how calls were labelled. These discontinuities arose due to a variety of human labellers for different sections of data and the presence of similar frequency components across different call types.

Tonal downsweep calls were excluded from the evaluation because labelled examples of this call type were only found in one of the three recordings.

Detections were evaluated on whether they overlap a labelled call, so the accuracy of the detected start and end times was not evaluated. Frequencies above 500 Hz were not considered, since the audio was downsampled to 1 kHz.

1.5 Layout of report

* Chapter 2 provides the background information needed to understand the problem, as well as literature on previous solutions to the problem.
* Chapter 3 provides background information on the dataset used for this project and explains how the data was split.
* Chapter 4 gives a detailed design of the entire project, including the feature extraction methods, the DTW implementation, hyperparameter tuning, the evaluation procedure and the evaluation metrics.
* Chapter 5 presents the results obtained from the designed system, as well as a description of what the results show.
* Chapter 6 contains the concluding thoughts on the results, including whether the objectives were met, as well as improvements and recommendations for the project.

Chapter 2

Literature

2.1 Bryde's Whales

Bryde's whales predominantly live in tropical or subtropical environments across the world. The different call types, rates, frequencies and harmonics of these whales vary depending on the geographical location of the colony. Generally Bryde's whales have calls containing frequencies from as low as 42 Hz to as high as 1 kHz [9].

In the dataset used for this project there were four main call types present. These are referred to as single-tones, multi-tones, burst-tonals and tonal downsweeps.

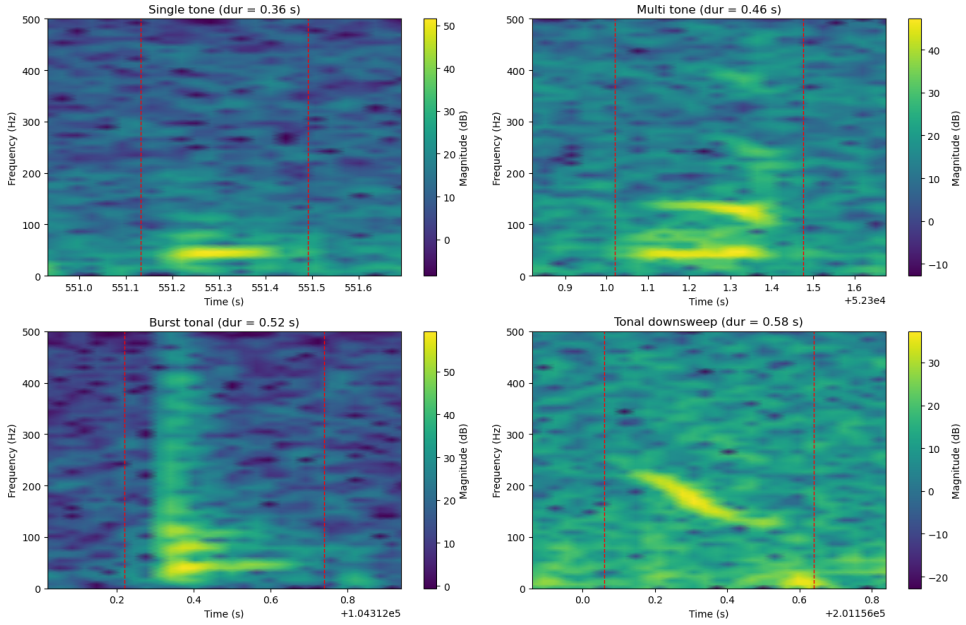


Figure 2.1: Examples of how each call type looks: single-tone, multi-tone, burst-tonal and tonal downsweep. The spectrograms for each call type were printed based on labelled calls from the dataset [1].

As explained by Faustmann et al. [3] The single-tone calls were characterised by a singular frequency at around 42 Hz. Multi-tones also contained the 42 Hz frequency, but there were one or two other frequencies above this with a maximum frequency component at about 200 Hz. Burst-tonals also have the multiple frequency characteristic of the multi-tones, however the lower frequencies, 42 - 200 Hz, were held for longer than the higher frequencies, 200 - 500 Hz. The tonal downsweep was the only call type not characterised by the 42 Hz frequency. It was a sweeping of frequencies through time starting at around 200 Hz and ending at around 100 Hz.

In accordance with the original detected calls the average call length of each call type was around 0.3 seconds with a maximum duration of about 2.5 seconds. In other studies of Bryde's whale calls in different geographical areas, a maximum duration of around 7.0 seconds was identified by Faustmann et al. [3] and Oleson et al. [9].

Decisions made in the project pertaining to frequencies and durations falling within the detection limits were made based on the above information.

2.2 Dynamic Time Warping

Dynamic time warping (DTW) is an algorithm for detecting the similarity of two signals, $X = [x_1, x_2, \ldots, x_n]$ and $Y = [y_1, y_2, \ldots, y_m]$. This measure of similarity is achieved by calculating an optimal alignment cost between the two signals of length $N \in \mathbb{N}$ and $M \in \mathbb{N}$ respectively. In terms of feature vectors such as the short-time Fourier transform (STFT), mel-frequency cepstral coefficients (MFCC) or linear predictive coding coefficients (LPCC) this means that each frame making up one signal is compared to each frame making up another signal. Using an alignment cost the best alignment for each frame in one signal to the other signal is chosen and an overall DTW cost is calculated to show how similar the overall signals are. Figure 2.2 shows, as explained by Müller and Zalkow [10] and Kamper [11], that such an optimal alignment path may mean that a single frame in signal X may produce the lowest cost against more than one frame in signal Y and vice versa.

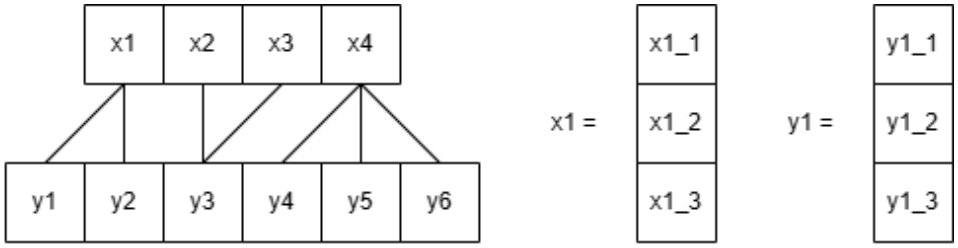


Figure 2.2: An example of the optimal warping path between two vector features, X with length N = 4 and Y with length M = 6

The overall cost needed to determine how similar the two signals are is calculated via a

cost matrix. Cost matrix, $C_{N+1,M+1}$ for signals X and Y, is initialised such that $C_{i,0} = \infty$ for i = 1 to N, $C_{0,j} = \infty$ for j = 1 to M and $C_{0,0} = 0$.

The formula used to fill each cell of the cost matrix is:

$$C_{i,j} = \text{dist}(x_i, y_j) + \text{Q} \qquad \text{with} \quad \text{Q} = \min \begin{cases} C_{i-1,j-1} & \text{(match)} \\ C_{i-1,j} & \text{(insertion)} \\ C_{i,j-1} & \text{(deletion)} \end{cases} \tag{2.1}$$

where $dist(x_i, y_j)$ represents the difference between X_i and Y_j as a distance. For vector signals this distance is calculated between each frame i of signal X and each frame j of signal Y. According to Kamper [11], this can be done using various different techniques including cosine, Euclidean or dot product similarity. For this project only the cosine distance and Euclidean distance metrics were used.

Euclidean distance is sensitive to the magnitude and direction of the vectors for which it is calculated. In basic terms, as described by Schwaber-Cohen [12], it is the absolute difference between two vectors.

$$d(\mathbf{x_i}, \mathbf{y_j}) = \sqrt{(\mathbf{x_i}_1 - \mathbf{y_j}_1)^2 + (\mathbf{x_i}_2 - \mathbf{y_j}_2)^2 + \cdots + (\mathbf{x_i}_n - \mathbf{y_j}_n)^2} \tag{2.2}$$

Cosine similarity is only sensitive to the direction of the vectors for which it is calculated. In basic terms, as described by Schwaber-Cohen [12], it is the angle between the two vectors.

$$\cos(\mathbf{x_i}, \mathbf{y_j}) = \frac{\mathbf{x_i} \cdot \mathbf{y_j}}{\|\mathbf{x_i}\| \, \|\mathbf{y_j}\|} \tag{2.3}$$

Since DTW minimises the cost between frames, the cosine similarity cannot be used directly as the distance, because a higher similarity must give a lower cost. Instead the cosine distance is used, which is defined in the SciPy documentation [13] as:

$$d_{\cos}(\mathbf{x_i}, \mathbf{y_j}) = 1 - \cos(\mathbf{x_i}, \mathbf{y_j}) \tag{2.4}$$

A cosine distance of 0 therefore means that the two vectors point in the same direction.

Q is the minimum value of three possible options namely match, insertion or deletion. Match is the value in the matrix cell diagonally down and left from the current cell for which the cost is being calculated. Insertion is the value in the matrix cell directly below the current cell and deletion is the matrix cell directly to the left of the current cell.

The cost which is then calculated for the current cell, $C_{i,j}$, is a summation of the distance between the feature frames i and j in signals X and Y respectively as well as the lowest accumulated cost up until that point. This means that the final cell reached within the cost matrix, $C_{N+1,M+1}$, will contain the lowest overall cost for the similarity of signals X and Y. The lower the final cost, the higher the similarity between the two signals.

The example shown in Figure 2.3 assumes that X and Y are scalar signals, however, as explained by Kamper [11], the same logic is used when X and Y are vectors.

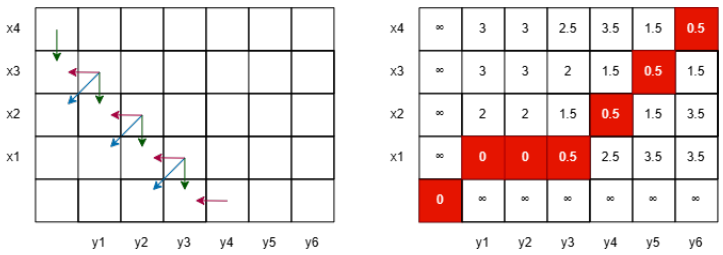


Figure 2.3: An example for the three possible options for min as well as an example of the optimal warp path found for signals X and Y assuming X=[0, 2, 1, 0] and Y=[0, 0, 0.5, 2, 1, 0]

The general case of DTW finds global alignment between the two signals. This alignment is found using the warping path as described by Alizadeh [14]. The warping path is the path of the best alignment between signals X and Y within the cost matrix, as shown in Figure 2.3. This path, W, is a sequence of matrix coordinates, (i, j), with a total length k.

The boundary condition for global alignment ensures the final path starts and ends with the start and end frames of each respective signal.

$$i_1 = 1, i_k = N \quad \text{and} \quad j_1 = 1, j_k = M \tag{2.5}$$

The monotonicity condition ensures that each new point in the warping path is a step forward in time from the previous step.

$$i_{t-1} \leq i_t \quad \text{and} \quad j_{t-1} \leq j_t \tag{2.6}$$

The continuity condition ensures the step size of the warping path is only one unit in time.

$$i_t - i_{t-1} \leq 1 \quad \text{and} \quad j_t - j_{t-1} \leq 1 \tag{2.7}$$

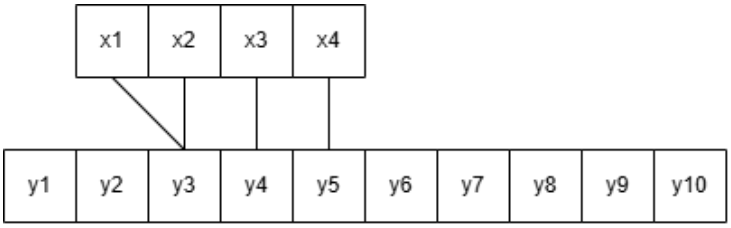


Figure 2.4: Example of a subsequence DTW alignment between two signals $X = [x_1, x_2, x_3, x_4]$ and $Y = [y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9, y_{10}]$

Subsequence DTW is a version of the DTW algorithm that alters the boundary condition such that global alignment is no longer necessary. Instead the entirety of signal X is now matched with a subsequence of signal Y. The new boundary condition allows the warping path to start and end at any point within the first and final row of the matrix respectively.

$$i_1 = 1,\ i_k = N \quad \text{and} \quad 1 \leq j_1 \leq j_k \leq M \tag{2.8}$$

In global alignment the final cell in the cost matrix, $C_{N+1,M+1}$, is considered the overall optimal cost for those two signals. As described by Müller and Zalkow [15], with subsequence DTW the entire final row of the matrix is checked to find the lowest cost. The position of that cost shows where the end point of the alignment detected via subsequence DTW is.

2.3 Feature Extraction Methods

2.3.1 Short-Time Fourier Transform

According to Usman et al. [16] and Kamper [17], the short-time Fourier transform (STFT) is used to analyse the change in frequency of a signal over time. The STFT of a segment of audio is calculated by multiplying the signal in the time domain, $x(t)$ with a window $w(t-\tau)$ which is swept across said signal. The Fourier transform (FT) of each window is calculated and used to determine the frequency information for that window. Thus, the STFT provides the change in frequency for a segment of audio throughout time.

The mathematical formulation of STFT is:

$$X(\tau, \omega) = \int_{-\infty}^{\infty} x(t)\, w(t-\tau)\, e^{-j\omega t}\, dt, \tag{2.9}$$

where τ is the position of the window as it shifts through time at step-size H and ω is the frequency being analysed.

The frequency resolution of the STFT is given as:

$$resolution = \frac{f_s}{L} \tag{2.10}$$

where f_s is the sampling frequency of the signal and L is the length of the window over which the FT is taken.

This means that the STFT has a trade-off between time and frequency resolution.

If L is kept short, time resolution is high since smaller segments of time are being used for one FT analysis. However, frequency resolution is then decreased since too little time passes for the FT to distinguish between frequencies spaced closer together.

If L is kept long, time resolution drops. This means that fast changes in frequency will be missed. However, frequency resolution increases and it becomes easier to distinguish between different frequency bands.

Window overlap is another consideration when using STFT. Increased overlap creates a smoother representation of the changes in frequency occurring over time since information at the edges of the window is no longer lost.

2.3.2 Mel-Frequency Cepstral Coefficients

Mel-frequency cepstral coefficients (MFCC) is a common choice for feature extraction when working on acoustic data. This is because, as explained by Kamper [18], it is set up to perceive frequencies as they are perceived by the human ear. That means the lower frequencies are given a higher resolution than the high frequencies. Features are initially extracted using the fast Fourier transform (FFT). After this they are passed through a mel-scale filter bank. The formula for the mel-frequency calculation given by Usman et al. [16] is:

$$Mel(f) = 1127 \ln\left(1 + \frac{f}{700}\right) \tag{2.11}$$

with $Mel(f)$ as the frequency value on the mel scale and f as the actual frequency in Hz.

This bank is made up of 20 to 40 triangular filters which are placed, smallest width to largest width according to the mel scale, along the frequency spectrum. According to Singh [19] and Kamper [18], the standard amount of triangular filters chosen is 26. This results in 26 coefficients for the entire range of frequencies being analysed.

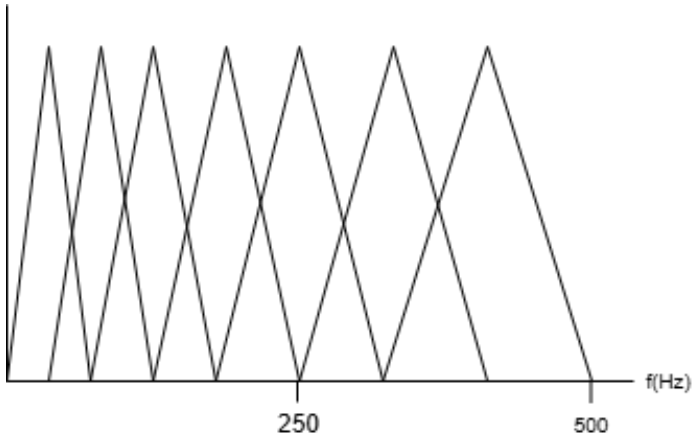


Figure 2.5: The mel-scale filter bank is made up of 20 to 40 triangular filters arranged according to the mel scale.

The log of these coefficients is then taken and finally the discrete cosine transform (DCT)

is used to decorrelate the filter bank coefficients. That means any patterns linking different coefficients to each other are removed such that the coefficients become independent. The final number of cepstral coefficients returned by the DCT is typically 12 or 13 since the most important information is kept there. The formula used by Usman et al. [16] to perform the DCT is:

$$MFCC_i = \sum_{k=1}^{K} X_k \cos\left(\frac{i(k-0.5)\pi}{K}\right), \quad i = 1, 2, \ldots, K \tag{2.12}$$

with i as the index of the current coefficient, K as the total number of triangular filters used, k as the specific filter bank at that point in time and X_k as the energy after log was applied to the mel coefficients.

As explained by DataMListic [20][21], the first and second derivatives, often referred to as deltas, of the cepstral coefficients are often used in feature extraction. They show the change of each cepstral coefficient across frames. This can be vital information when analysing how frequency changes across time since the MFCC process tends to lose this information. The formula for the derivatives, as given by Majeed et al. [22], is:

$$x_t^{(d)} = \frac{\sum_{m=-M}^{M} m \cdot x_{t+m}^{(d-1)}}{\sum_{m=-M}^{M} m^2} \tag{2.13}$$

with t as the specific point in time, d as the order of derivative, M as the number of frames on either side of t and m as the frame offset.

2.3.3 Linear Predictive Coding Coefficients

As explained by Rabiner and Juang [23] and Usman et al. [16], linear predictive coding (LPC) is a straightforward mathematical method commonly used for feature extraction. LPC focuses on the method of approximating a sample of a speech signal using a combination of p samples from the past. This means that for each sample for which there are p available samples before it an approximation is made in the form of the equation:

$$\tilde{s}(n) = \alpha_1 s(n-1) + \alpha_2 s(n-2) + \cdots + \alpha_p s(n-p) \tag{2.14}$$

where $\tilde{s}(n)$ is the approximated speech sample and $\alpha_1, \alpha_2, \ldots, \alpha_p$ are the coefficients for that approximation. The p coefficients used are constant over all samples within the given frame.

For each sample with an approximation an error is calculated between the real sample and the approximated sample.

$$e_n = s(n) - \tilde{s}(n) \tag{2.15}$$

The sum of the squares of all these errors is minimised to find the p constant coefficients.

The equation for this is:

$$E_n = \sum_m e_n^2(m) \tag{2.16}$$

where m is the sample for which the error $e_n(m)$ was calculated.

The feature extracted via this method is made up of the p LPC coefficients which were calculated to minimise the mean square error. These coefficients tend to be unstable, therefore cepstral coefficients can be calculated from these raw LPC coefficients giving linear predictive coding coefficients (LPCC). LPCC gives the coefficients of the Fourier transform (FT) representation of the magnitude spectrum and is more reliable than the LPC representation.

LPCC can be calculated from the LPC results through recursion. The formula for that is:

$$\begin{aligned} c_0 &= \ln \sigma^2 \\ c_m &= a_m + \sum_{k=1}^{m-1} \left(\frac{k}{m}\right) c_k a_{m-k}, \qquad 1 \leq m \leq p \\ c_m &= \sum_{k=1}^{m-1} \left(\frac{k}{m}\right) c_k a_{m-k}, \qquad m > p, \end{aligned} \tag{2.17}$$

where σ^2 is the LPC gain term.

The final feature vector extracted using LPCC contains p coefficients for a single frame, with the zeroth coefficient, c_0, containing the magnitude information.

2.4 Similar Studies

2.4.1 Classification of Killer Whale Vocalisations using DTW

Brown and Miller evaluated four different variations of the DTW algorithm to classify biphonic killer whale vocalisations using fundamental frequency contours (see [24]).

Due to the biphonic nature of these calls the low-frequency component and high-frequency component for each call was extracted for separate analysis. Four different variations of the DTW algorithm with various different constraints were used to evaluate against each other.

The final result of the paper was that the method referred to as “Sakoe-Chiba” on the LPC performed the best with a 90% agreement with the labels selected by human classifiers.

This study did show that the use of DTW in bioacoustic analysis is a good option. However, it did not consider the ability of DTW to detect calls in a continuous segment of

bioacoustic data as all calls which were classified had already been selected and segmented to the pitch contours of the vocalisation. This study also did not consider the impact of low frequency analysis using DTW as killer whale vocalisations do not tend to fall below 500 Hz.

2.4.2 Analysis of Template-Based Detection Algorithms for Bryde's Whales

Ogundile and Versfeld analysed two template-based detection algorithms to detect short pulse calls for Bryde's whales in False Bay, South Africa (see [25]). The two methods tested were dynamic time warping (DTW) and linear predictive coding (LPC). In their analysis they used two separate sample sets with different signal-to-noise ratio (SNR) values to also evaluate the performance of these techniques in the presence of noise. The input features for these models were the time-domain representation of the signals.

The final results of the article found that both LPC and DTW performed well in high and low SNR segments. LPC performed the overall best in low noise segments.

This article provides insight into the use of template-based detection systems for Bryde's whales. This is relevant since DTW which is handled in this article is the chosen detection mechanism for this project. While LPC is also used in the article, its use is vastly different to the use within this project. LPC is used as a detection algorithm and tested on its ability to act as such. In this project LPC is not used as a detection algorithm, but instead linear predictive coding coefficients (LPCC) is used as a feature extraction method. The detection method focuses on the error component of the LPC equation while the feature extraction method focuses on the coefficients calculated for the approximation of the real signal.

2.4.3 Detection of real-time fin and blue whale calls

Mouy et al. [26] tested detection models on real-time data for fin and blue whales. The detection methods used were dynamic time warping (DTW) and vector quantisation (VQ).

Instantaneous frequency, speed and acceleration were extracted as features. It was found that due to low frequency components from background elements such as ships the detection of false positives (FP) increased.

In terms of this project the article shows how low frequency detections can be affected by low frequency background noise. This article also considered real-time audio detection which is much closer in similarity to detection within a large audio file such as what was done in this project. This article did not use the same feature vectors as the project and did not consider their effect on DTW performance.

2.4.4 Hidden Markov Model for Bryde's Whale short pulse calls

Ogundile et al. [27] compared a time domain feature extraction technique used for hidden Markov models (HMM) against mel-frequency cepstral coefficients (MFCC) and linear predictive coding (LPC), frequency domain techniques. The model is used for detection of Bryde's whale calls in False Bay, South Africa.

These three HMM models are used on 2 hours of continuous recordings to detect these calls. This is similar to the continuous recording detection done in this project.

The time domain feature HMM model performed the best with MFCC performing second best and LPC performing the worst. It was found that the MFCC and LPC models were sensitive to background noise.

While this does test detection on a continuous segment of audio, this article focuses on HMM detection and not DTW. Similar to this project it compares the MFCC feature extraction method with the LPC method. However, considering the fact that the LPC features were not converted to linear predictive coding coefficients (LPCC) this does not fully align with this project.

2.4.5 Speech Recognition using DTW

Permanasari et al. [28] analysed the effectiveness of dynamic time warping (DTW) in matching speech patterns. Variabilities were included in the study via pronunciation timing, volume and speaker identity.

Features were extracted from the recorded audio. Some audio was used as templates and others as test data. DTW was then used to test which of the variations impacted DTW distances the most. The main tests were keeping the pronunciation time the same with the same volume, different pronunciation time with the same volume and same pronunciation time with different volume.

It was determined that DTW was an effective pattern recognition algorithm and that pronunciation timing had a larger impact on DTW alignment than volume.

This study also uses DTW to recognise patterns in audio, however this focuses on speech recognition and the effects of variability on that recognition. It does not focus on different feature vectors or detection of low frequency whale calls in a continuous recording.

2.5 Chapter Summary

In this chapter, Bryde's whale traits were discussed, the mathematics and theoretical explanations for DTW, STFT, MFCC and LPCC were given and various literature pieces were discussed with similar methodologies or goals. In the next chapter details of the

dataset used for the project will be explained.

Chapter 3

Dataset

3.1 Background

The dataset used contains three 72-hour recordings, recorded in False Bay, near Gordon's Bay, between 22 April 2023 and 22 June 2023. The recordings were taken at a depth of 20 to 30 metres at the coordinates S 34.17°, E 18.76°.

The device used for the recordings was a single omnidirectional hydrophone, designed and built by a research group in the Department of Electrical and Electronic Engineering at Stellenbosch University.

The recorded data has a sampling rate of 8 kHz and a 16-bit resolution. The data was extracted as a 72-hour Waveform Audio File Format (WAV) file from an Secure Digital (SD) card used by the recording device [29].

3.2 Labelling

Detection tables, containing the start and end times of calls within each of the three files, were provided by a previous research project that used a band-limited energy detector to flag calls. This method unfortunately did not detect all the calls within the audio files, and none of the calls were labelled as a specific call type [29].

Another detection table was obtained from Tonitz et al. [1], which contained more detections for the first 72-hour file and call labels for all three files. This meant that, while all three files contained labelled calls, only one contained enough labelled calls to use the entire file without expecting an excessive number of unlabelled calls.

3.3 Data Splitting

For this project the data was split into three sections: template data, hyperparameter tuning data and evaluation data. Each of the 72-hour audio files was assigned to a different section. Given that only the first of the three 72-hour audio files was well labelled, it was decided that this file would be used as the evaluation data. The second audio file was used for hyperparameter tuning and the third was used to select templates. Labelled calls for the tonal downsweep call type were only found in the audio file used for evaluation. Therefore, due to a lack of data, this call type was ignored.

Table 3.1: Dataset counts by date and category

Recording Start Date	Call Count				Total Calls
	Single-Tone	Multi-Tone	Burst-Tonal	Tonal Downsweep	
22 April 2023	1015	452	171	164	1802
26 April 2023	242	91	9	0	342
29 April 2023	32	25	11	0	68

3.3.1 Template Selection

The audio file recorded starting on 29 April 2023 was used to select templates. These templates were used as the baseline for DTW comparisons.

For the three considered call types (single-tones, multi-tones and burst-tonals), the maximum number of templates per call type that could be extracted, such that each call type had the same number of templates, was 10. The limiting call type was burst-tonals. In Table 3.1 it can be seen that there were 11 available burst-tonals. However, it was decided to make 10 the maximum number of templates per call type.

To ensure that the 10 templates chosen were representative of varying SNR values, templates were chosen by looking at spectrograms of labelled calls throughout the audio file. The visual inspection was also necessary since some of the calls labelled as single-tones looked more like multi-tones and some of the multi-tones could actually be classified as burst-tonals. This overlap in possible classification was due to the variety of human labellers and the common frequency components between call types.

3.3.2 Hyperparameter Tuning Template Selection

The audio file recorded starting on 26 April 2023 was used to select templates used for hyperparameter tuning, referred to as call type calibration templates. These call type calibration templates were used to determine the DTW cost threshold below which a

segment of audio could be considered a call. For the call type calibration templates, 10 templates were chosen for each call type. This was again limited by the maximum number of burst-tonal templates. In this file there were only 9 labelled burst-tonals. However, when selecting multi-tone templates, one call originally labelled as a multi-tone was judged, based on its spectrogram, to be a burst-tonal. This call was relabelled and used as the tenth burst-tonal template, so that each call type had 10 call type calibration templates. The call counts in Table 3.1 reflect the original labels, before this change.

Besides these 30 call type calibration templates (10 for each call type), 100 background calibration templates were also selected. Within each audio file, most of the time was made up of background noise and only some parts actually contained calls. This was simulated by selecting many more background calibration templates than call type calibration templates. Since there is no table labelling background noise, background calibration templates of varying sizes were extracted from the file and 100 templates were manually selected. The reason for the manual selection process is linked to the unreliability of the labelled data. The audio file used for selection of calibration templates was only labelled by the band-limited energy detector and thus did not label all the calls within the file. This means that some of the samples extracted as background audio actually contained calls that had never been labelled. Manually selecting each template ensured that all the templates chosen contained only background noise.

3.3.3 Evaluation Data Selection

The audio file recorded starting on 22 April 2023 was used for the final evaluation of the whale call detection model. No templates were extracted; instead, the entire file was swept to search for calls.

As seen in Table 3.1, this file was the only file that contained tonal downsweep calls. Due to the limited number of these calls, all the labelled tonal downsweeps within this file were ignored while testing.

3.4 Chapter Summary

In this chapter the background for the data used in this project was discussed and the method used for splitting the data was explained. In the next chapter the design and implementation of the Bryde's whale call detection system was explained.

Chapter 4

System Design

4.1 Overview

The project was coded in Python. The functionality of each section of the project was placed in relevant Python scripts. These were imported into the relevant Jupyter notebook used for each section.

4.2 Feature Extraction Methods

The raw audio recorded by the hydrophone was stored in WAV files with a sampling frequency of 8 kHz. Before extracting feature vectors from the recordings, the audio was first downsampled to a 1 kHz sampling rate. This helped remove unnecessary frequency information, as Meddins [30] explains that the Nyquist frequency ($\frac{f_s}{2}$) for a 1 kHz sampling rate is 500 Hz.

Faustmann et al. [3] report that most Bryde's whale vocalisations occur below 500 Hz. Thus, this sampling rate also ensures that no relevant information was lost prior to analysis.

4.2.1 Short-Time Fourier Transform

The `scipy.signal.stft` function from the SciPy library was used to calculate the STFT feature vectors. The window length L, as referenced in Eq. (2.10), is calculated by multiplying f_s, the sampling frequency of the given audio segment, by the chosen STFT frame duration. No fixed frame duration was pre-selected, as the optimal frame duration was determined during the hyperparameter selection stage of this project.

The window overlap H, as referenced in Eq. (2.10), is calculated by multiplying L by the chosen overlap percentage. For this project, the overlap percentage was set to 0.75. This high overlap percentage produced a smooth spectrogram when viewing the time-frequency representation of a given audio segment. A Hamming window was chosen

to reduce spectral leakage caused by the finite length of the windows used to extract the feature vectors, as described by Kamper [17] and in the windowing functions guide [31].

4.2.2 Mel-Frequency Cepstral Coefficients

The `librosa.feature.melspectrogram` and `librosa.feature.mfcc` functions from the Librosa library, as documented by the Librosa development team [32], were used to calculate the cepstral coefficients for the MFCC feature extraction implementation. The window length required for the FT is calculated by multiplying f_s by the chosen frame duration, which is optimally selected during hyperparameter tuning. Unlike the STFT function, `librosa.feature.mfcc` requires the hop length rather than the overlap length. The hop length is calculated by subtracting the window overlap, H, from the frame length, L. Here, H is calculated by setting the overlap percentage to 0.75 and multiplying it by L. A Hamming window is used to reduce spectral leakage.

The Librosa function used to calculate the MFCC features automatically applies DCT type 2 and ortho-normalisation, as described in the SciPy documentation [33]. These settings mean that the bottom row of the output matrix, c_0, represents the sum of the log-spectrum energy. Roch et al. [34] note that it is common practice in speech analysis to remove coefficient 0, as it contains the total signal energy and can dominate distance calculations in subsequent analysis. For this project, only coefficients 1 to 14 were retained. This ensured that the analysis considered only the overall shape of the log spectrum and not the total energy of the signal. The number of mel-spectrogram filters was set to 26, which is standard practice in audio processing. A total of 14 final coefficients were extracted (since 13 are typically used, retaining 14 allows 13 coefficients to remain after removing c_0).

The `librosa.feature.delta` function, as documented by the Librosa development team [35] and described by McFee et al. [36], was used to calculate the first and second derivatives for the MFCC features. The feature vector returned when using the first derivative was a concatenation of the original MFCC feature vector with its first derivative. This resulted in a feature vector with 26 coefficients (the 13 original coefficients plus 13 derivative coefficients). Similarly, for the second derivative, the concatenation of the original MFCC and first derivative was further concatenated with the second derivative, yielding a 39-coefficient feature vector.

4.2.3 Linear Predictive Coding Coefficients

The `librosa.util.frame` function [37] and the `librosa.lpc` function [38] from the Librosa library were used to calculate the LPC features. The STFT and MFCC functions frame the audio internally; therefore, framing prior to feature extraction was not required for those methods. However, the LPC function accepts a sequence of pre-framed audio

segments to calculate the approximation coefficients. The `librosa.util.frame` function was used with a frame duration selected during hyperparameter tuning and a hop length of 0.25 of the frame duration $(1 - 0.75)$. A Hamming window was applied to the frames before calling the `librosa.lpc` function.

Rabiner and Juang [23] explain that a higher LPC order yields a more detailed spectral approximation. However, considering that background noise could adversely affect coefficient calculations if the model becomes overly detailed, an order of $p = 4$ was selected.

Due to the numerical instability of raw LPC parameters, the LPC features were converted to LPCC. This was implemented using the recursive formula in Eq. (2.17). According to Rabiner and Juang [23], the number of LPCC coefficients typically selected is approximately $\frac{3}{2}p$. For $p = 4$, approximately 6 LPCC coefficients would standardly be calculated. After testing different numbers of coefficients, 8 LPCC coefficients were found to yield the best f1-scores during hyperparameter selection. Therefore, 8 LPCC coefficients were used.

4.3 DTW Methodology

The `librosa.sequence.dtw` function from the Librosa library, as documented by the Librosa development team [39], was used to calculate the DTW cost matrix between a template and an audio segment. The pairwise distance matrix required as input to this function was computed using the `scipy.spatial.distance.cdist` function from the SciPy library, as described in the SciPy documentation [13].

The distance function received the template feature and the audio segment feature extracted via STFT, MFCC, or LPCC, along with the specified distance metric. The two distance metrics evaluated were cosine distance and Euclidean distance. As shown in Eq. (2.3) and Eq. (2.2), cosine similarity calculates the angular difference between two vectors, whereas Euclidean distance calculates the straight-line distance between them. For the STFT feature extraction method, cosine distance was selected rather than Euclidean distance because cosine distance is invariant to magnitude. Consequently, the distance between a template and a distant (softer) call versus a nearby (louder) call depends solely on spectral shape rather than signal amplitude. This ensured that calls of varying loudness could be matched based on spectral properties.

For the MFCC implementation, signal energy variations due to distance from the recording device were mitigated by removing c_0, as described in Section 4.2.2. Similarly, for LPCC, signal energy is captured by the zeroth coefficient c_0, which is fixed to 1 during LPC extraction and excluded from the 4 primary LPC coefficients. As a result, Euclidean distance could be used without interference from signal amplitude, as only the spectral envelope remains. Because Euclidean distance is standard for DTW applications and cosine distance provided no extra benefit for amplitude-normalised features, Euclidean

distance was chosen for MFCC and LPCC.

The `subseq` parameter in `librosa.sequence.dtw` was set to `True`. As described in Section 2.2, subsequence DTW relaxes boundary constraints so that the start and end points of a detected match do not need to align with the boundaries of the candidate sequence. When scanning 1-hour segments of a 72-hour audio file, subsequence DTW enables the detection of multiple calls within a single segment by identifying dips along the final row of the cost matrix and selecting the local minima as call end points. This is detailed further in Section 4.5.

4.4 Hyperparameter Tuning

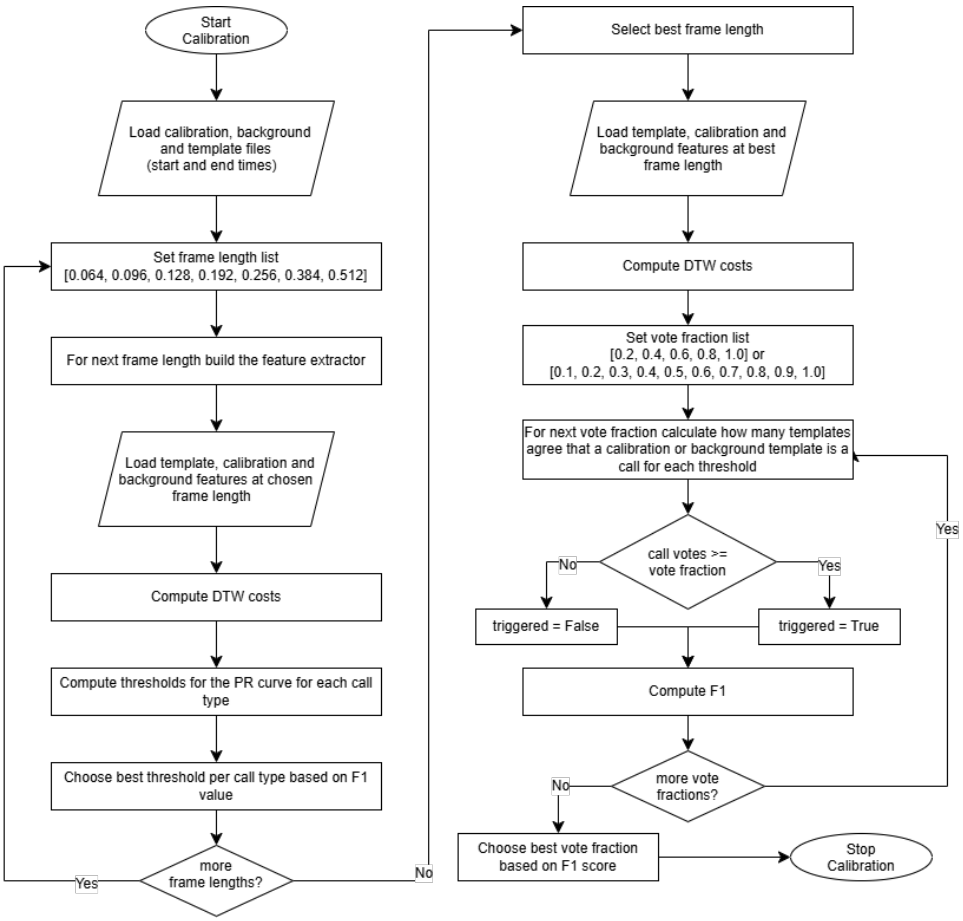


Figure 4.1: Software flow diagram describing the hyperparameter selection procedure.

Three hyperparameters were selected for tuning: feature frame length, call type cost thresholds, and the call vote fraction. Figure 4.1 illustrates the overall execution flow for hyperparameter selection.

As defined in Eq. (2.10), frequency resolution is given by frequency resolution $= \frac{f_s}{L}$. Because Bryde's whale single-tone calls primarily feature a fundamental frequency around

42 Hz and multi-tone calls contain components between 42 and 200 Hz, an upper bound on frequency resolution was set to approximately 10 Hz, corresponding to a frame duration of 100 ms. The lower frequency resolution bound was set to approximately 2 Hz. Although a finer frequency resolution improves spectral bin separation, a 2 Hz resolution requires a frame duration of 500 ms. At a 75% window overlap, this leaves a hop length of 128 ms. Given that the average call duration in the calibration dataset is 246 ms, a 128 ms hop length remains sufficiently short to resolve individual calls. Hees [40] explains that FFT routines are most computationally efficient for lengths that are powers of two. To maintain computational efficiency, candidate frame lengths were therefore selected as powers of two or multiples thereof: $(0.096, 0.128, 0.192, 0.384, 0.512)$ seconds.

Because the call type and background calibration templates had to be manually selected due to data scarcity, each calibration template was extended on both sides. This extension ensured that subsequence DTW operated under conditions representative of the evaluation pipeline despite dataset limitations. The total extended length was configured to exceed the longest evaluation template by at least one frame length on either side. This extra margin was chosen to match the maximum tested frame duration (0.512 s). Each call or background calibration template was centred within this window by adjusting its boundary timestamps accordingly.

For each tested frame length, three decision thresholds were determined (one per call type). These thresholds were determined by evaluating DTW costs between evaluation templates and call type calibration templates, as well as between evaluation templates and background calibration templates. The resulting costs were grouped into call-calibration costs and background-calibration costs. Two hundred candidate thresholds were spaced linearly between the overall minimum and maximum costs across both lists. The f1-scores were evaluated across all candidate thresholds for each call type as described in Section 4.6. The threshold yielding the highest f1-score was selected for each respective call type.

Both call type calibration templates and background calibration templates were used to determine the thresholds since true positive (TP), false positive (FP) and true negative (TN) calls were all needed in the calculation of precision and recall and thus in the calculation for the f1-score used to select the optimal thresholds. This can be seen in Eq.4.1, 4.2 and 4.3.

For a given frame length, the minimum f1-score among the three call type thresholds was recorded. The optimal feature frame length was then selected as the one whose minimum threshold f1-score was highest across all tested frame lengths.

Although three distinct thresholds were computed (one per call type), this project focuses exclusively on general call detection rather than classification. This approach was adopted because significant spectral overlap exists among call types. Furthermore, label inconsistencies (arising from differing annotations across human labellers) meant that templates selected for a call type might differ from the ground-truth definitions used in

evaluation data. Separate thresholds per call type were nevertheless maintained to account for structural acoustic variations during candidate detection.

In the evaluation pipeline, candidate audio segments are compared against sets of 5 or 10 reference templates per call type. Because templates capture minor structural variations within a call type, individual templates may disagree on whether a candidate segment contains a call. A vote fraction parameter was introduced such that a candidate segment is classified as a call only if a specified fraction of per call type reference templates trigger a detection. Candidate vote fractions were evaluated on the optimal frame length. For 5-template configurations, tested vote fractions were $\{0.2, 0.4, 0.6, 0.8, 1.0\}$; for 10-template configurations, tested fractions were $\{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$.

For each vote fraction, the overall f1-score for general call detection was computed across all call type calibration and background calibration templates. This meant that each group of 5 or 10 templates per call type and its relevant threshold was tested against all call type and background calibration templates. The vote fraction producing the highest overall f1-score was selected.

4.5 Evaluation Procedure

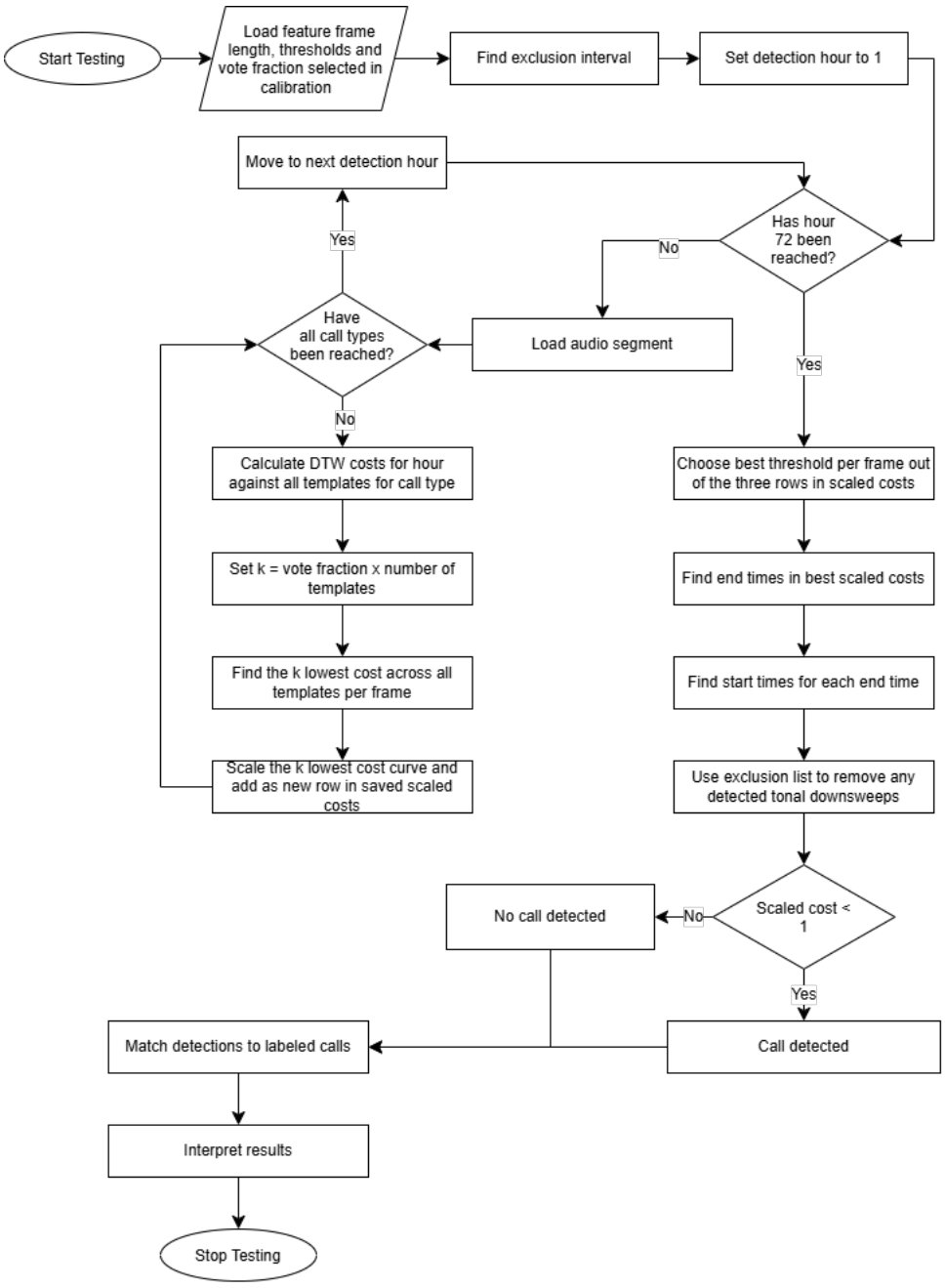


Figure 4.2: Software flow diagram describing the evaluation procedure.

4.5.1 Detection Methodology

The parameters selected during hyperparameter tuning (frame duration, call type thresholds, and vote fraction) were applied directly during evaluation. The evaluation dataset was the only recording containing tonal downsweeps. Because no calibration threshold was computed for tonal downsweeps, an exclusion index list was maintained for all annotated

tonal downsweeps. To avoid introducing discontinuities in time into the subsequence DTW cost matrix, tonal downsweeps were only excluded after detection rather than omitted prior to processing.

The 72-hour evaluation recording was processed in sequential 1-hour segments to maintain a low memory footprint. The `soundfile` Python library, as documented by Bechtold [41], was used to stream continuous hourly audio blocks without loading the entire dataset into memory. Each hourly segment was downsampled from 8 kHz to 1 kHz using the `scipy.signal.resample_poly` function described in the SciPy documentation [42], and features were extracted as outlined in Section 4.2.1.

All templates (5 or 10 per call type) were tested against the entire 1-hour audio segment. End frames of detected calls were identified if the percentage of templates for a specific call type detecting a call was equal to the vote fraction chosen during hyperparameter selection. This is explained below with the help of Figure 4.3, 4.4 and 4.5.

The DTW cost matrix was computed for all templates across each call type, and the bottom row of each cost matrix was extracted and stored, as illustrated in Figure 4.3.

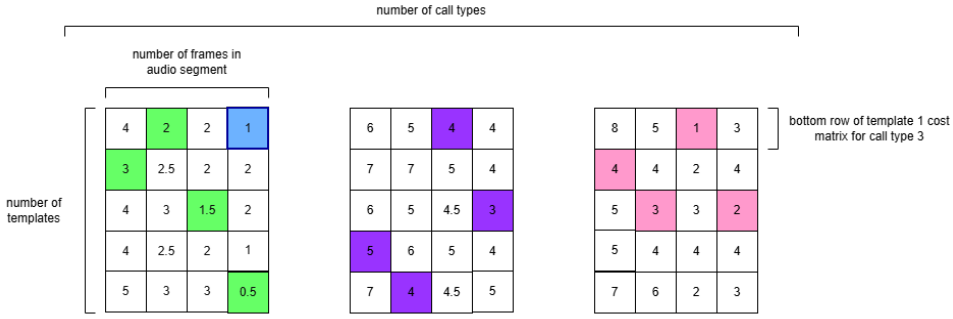


Figure 4.3: Illustration of per call type matrices containing the bottom row of the cost matrix for each template in the given call type for k = 1. The green, purple and pink blocks are the k_{th} lowest cost per frame for each matrix. These correspond to Figure 4.4 The blue block is the detected end frame as it appears in the template analysed to find the start frame. This corresponds to Figure 4.5.

The integer parameter k was defined as the vote fraction multiplied by the number of templates per call type. For each frame in the 1-hour audio segment, the k_{th} lowest cost across templates per call type was extracted into a single array per call type. The three k_{th} lowest costs arrays (one for each call type) were stacked to form a three row k_{th} best costs matrix as shown in Figure 4.4.



Figure 4.4: Illustration of the k_{th} best costs matrix. The coloured blocks correspond to those in Figure4.3

All the costs in the rows of the matrix were divided by the call type threshold for each row. This normalisation created a matrix of scaled k_{th} costs, where a value less than or equal to 1.0 consistently indicated a valid threshold match across all call types.

The minimum scaled cost across all three call types was taken per frame and placed in a cost per frame array. This array was evaluated for call detections (top array in Figure 4.5).

Evaluating the k_{th} lowest cost guarantees that the vote fraction requirement is satisfied: if the k_{th} cost falls below the decision threshold, at least k templates have matched the candidate segment.

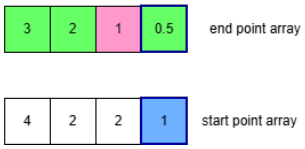


Figure 4.5: Illustration of the cost arrays used to find the start and end frames without normalisation. The coloured blocks corresponds to those in Figure 4.3 and 4.4.

Call end times were detected using the `scipy.signal.find_peaks` function, as described in the SciPy documentation [43], on the inverted cost array. In subsequence DTW, candidate matches correspond to local minima along the final row of the accumulated cost matrix. Inverting the cost array ($-$cost) transforms these local cost dips into peaks, allowing candidate end frames with cost ≤ 1.0 to be identified using standard peak-finding algorithms.

For each detected end frame, the corresponding call start time was determined. While the end time was identified via the lowest k_{th} lowest cost among all call type templates, the start time was extracted using the template yielding the absolute lowest cost (from the call type templates of the k_{th} lowest cost call type) at that end frame. Bottom array in Figure 4.5 shows the array selected to find the start time based on a detected end frame. The costs in the selected start time array were also normalised by dividing by the threshold for the given call type.

A 10-second audio window preceding the detected end frame was extracted for realignment. Because subsequence matches can exceed the baseline template duration, a

10-second window was chosen to accommodate long calls without requiring full recalculation across the entire 1-hour block.

The `librosa.sequence.dtw` function was re-executed on this localized window to compute the step matrix. Backtracking was then performed via `librosa.sequence.dtw_backtracking`, forcing the warp path to originate at the chosen end frame and trace backward to determine the exact start frame. Using the best-matching template (rather than the k-th template) ensures the most accurate start-time boundary estimation.

4.5.2 Matching Methodology

Detected calls were evaluated against labelled calls within the corresponding time period. The `numpy.searchsorted` function, as documented by the NumPy developers [44], was used to find the labelled calls overlapping the detected calls. This was achieved efficiently using four 1D arrays: detected start times, detected end times, labelled start times, and labelled end times.

For each detected call, `numpy.searchsorted` identified the index `lo` in the labelled end-time array where the detected start time would fit sequentially. Similarly, it identified the index `hi` in the labelled start-time array where the detected end time would fit. A condition of `hi` $>$ `lo` indicates that at least one labelled call overlaps with the detected window. If this condition was met, the detection was categorised as a true positive (TP), and the corresponding labelled call was marked as matched. Subsequent detections overlapping an already-matched label were still treated as TPs, as the primary objective of this system is detection rather than specific matches of start and end times.

4.6 Evaluation Metrics

Although receiver operating characteristic (ROC) analysis, as explained by Fawcett [45], is widely used in detection tasks, precision-recall (PR) analysis was chosen due to severe class imbalance between whale calls and background noise. Saito and Rehmsmeier [46] show that ROC curves can present an overly optimistic assessment in heavily imbalanced scenarios, whereas PR curves more accurately reflect performance under class skew. Consequently, precision, recall, and f1-score were selected as primary evaluation metrics. However, because the calibration dataset exhibited a different class imbalance ratio than the evaluation dataset, the calibrated operating point was expected to sit slightly off the theoretical optimum of the evaluation PR curve.

Precision measures the proportion of detected calls that correspond to true labelled calls. A true positive (TP) detection is defined as a detected call window overlapping at least one labelled call, while a false positive (FP) is a detection with no label overlap:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{4.1}$$

Recall measures the proportion of labelled calls successfully detected by the system. In this context, a TP label is defined as a labelled call overlapped by at least one detection, whereas a false negative (FN) is a labelled call that receives no detection overlap:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{4.2}$$

Note that TP definitions differ slightly between precision and recall. Because a single long labelled call may produce multiple detections (or vice versa), any overlap between call detections and labelled calls is treated as a true positive, supporting the system's objective of general call detection.

The f1-score represents the optimal mean of precision and recall:

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4.3}$$

Hyperparameter selections and model comparisons were evaluated primarily using the maximum f1-score. Additionally, average precision (AP) was calculated across the continuous PR curve using all-point interpolation, as defined by Padilla et al. [47]:

$$\text{AP}_{\text{all}} = \sum_{n} (R_{n+1} - R_n) P_{\text{interp}}(R_{n+1}) \tag{4.4}$$

where

$$P_{\text{interp}}(R_{n+1}) = \max_{\tilde{R} \geq R_{n+1}} P(\tilde{R}) \tag{4.5}$$

To construct the continuous PR curve for AP calculations, the decision thresholds were swept across a range using a scaling factor α. During initial processing, candidate end points were retained if their scaled cost was below 1.5. A strict cut-off at 1.0 was applied for single-point baseline matching, while retaining detections up to 1.5 ensured sufficient candidate points to generate a full PR curve over 100 linearly spaced threshold steps between 0.01 and 1.5.

4.7 Chapter Summary

In this chapter the design of all sub-systems built and combined to form the Bryde's whale call detection system was explained (Feature extraction and DTW implementations, Hyperparameter selection implementation and evaluation procedure). In the next chapter the results for all experiments conducted on the system will be discussed.

Chapter 5

Results

After conducting experiments for the three different models (STFT, MFCC and LPCC), analysis must be done to determine which of the feature extraction methods worked best for dynamic time warping (DTW). This analysis will be done by comparing f1-scores and average precision (AP).

5.1 Experiment 1: STFT Features

For the short-time Fourier transform (STFT) model used for call detection, two experiments were conducted. The first considered the results given the use of 5 templates per call type as the analysis database for both hyperparameter selection and evaluation. The second considered the use of 10 templates per call type, the original 5 with 5 new templates added to increase the database diversity.

Table 5.1: Calibration results for the STFT model: f1-scores and thresholds per call type, average f1-score and selected frame length (in seconds) for 5 and 10 templates per call type.

Number of Templates	Single Tones		Multi Tones		Burst Tonals		Average F1	Frame Length
	F_1	Threshold	F_1	Threshold	F_1	Threshold		
5	0.682	0.234	0.568	0.256	0.796	0.260	0.682	0.512
10	0.690	0.245	0.663	0.236	0.697	0.230	0.683	0.512

There was no real difference between the average f1-scores for the 5- and 10-template experiments during threshold selection, 0.682 and 0.683. Both experiments chose 0.512 seconds as the best frame length. This can be seen in Table 5.1. For both the 5- and 10-template experiments, the lowest f1-score of the three thresholds was for multi-tones, 0.568 and 0.663. The second lowest was for the single-tones, 0.682 and 0.690, and the highest was for burst-tonals, 0.796 and 0.697.

The similar average f1-scores for the two experiments show that the diversification of templates does not have an effect on threshold selection. The lowest f1-scores point to the possible effect of multi-tones often being mixed with both single-tones and burst-tonals because of human error incorporated by different labellers.

Table 5.2: Vote fraction selection results for the STFT model: precision, recall and f1-score at the chosen vote fraction for 5 and 10 templates per call type.

Number of Templates	Precision	Recall	F1	Vote Fraction
5	0.964	0.900	0.931	0.6
10	0.967	0.967	0.967	0.4

The precision, recall and f1-scores for the two STFT experiments during vote fraction selection can be seen in Table 5.2. The vote fraction for the 5-template experiment was 0.6 and for the 10-template experiment it was 0.4. This means that 3 of 5 templates and 4 of 10 templates had to be selected for each experiment.

In terms of the precision, recall and f1-score, the 10-template experiment performed the best for its chosen vote fraction. However, both experiments had high scores above 0.9 for each measure and therefore both performed well. There is an expected drop in precision, recall and f1-score for detection of evaluation data. This drop is due to thresholds being tuned on the data used for testing of vote fractions. Therefore, these results do not evaluate the overall performance of the model. However, they do show that diversification of templates did not affect the effectiveness of vote fraction selection. This can also be seen in that a similar number of templates needed to agree on a call type for detection to occur in both experiments.

Table 5.3: Final evaluation results for the STFT model: precision, recall, f1-score and average precision (AP) for 5 and 10 templates per call type.

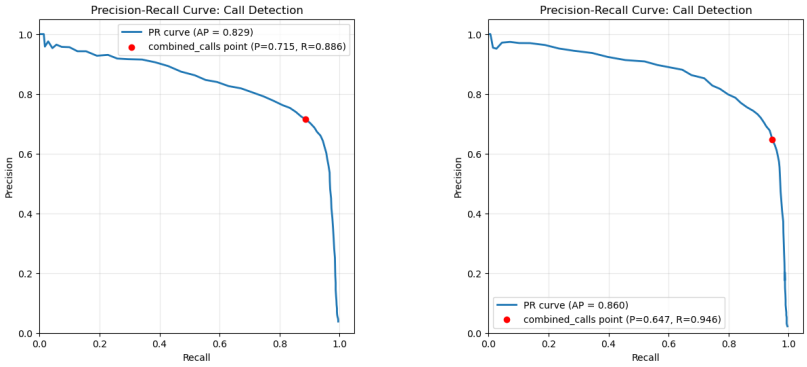
Number of Templates	Precision	Recall	F1	Average Precision
5	0.715	0.886	0.792	0.829
10	0.647	0.946	0.769	0.860

In terms of precision, as seen in Table 5.3, the 5-template experiment performed the best with a precision of 0.715 compared to 0.647 for the 10-template experiment. However, in terms of recall the 10-template experiment performed the best with a recall of 0.946 compared to 0.886 for the 5-template experiment.

For both experiments precision was substantially lower than recall. This shows that the thresholds selected were more lenient, therefore letting more background segments be classified as detected calls.

An increase in template diversity appeared to negatively affect the leniency of the model.

By looking at Figures 5.1a and 5.1b, it can be seen that both experiments performed well in terms of the selection of the hyperparameters, as both marked precision and recall points are in the optimal position of their respective curve. Therefore, despite the leniency of the model, both STFT experiments performed well in terms of hyperparameter selection.



(a) 5 templates **(b)** 10 templates

Figure 5.1: The PR curves for STFT

By again looking at Table 5.3 and Figures 5.1a and 5.1b, it can be seen that in terms of average precision (AP), and therefore the overall precision-recall (PR) curve of the model, the 10-template experiment outperformed the 5-template experiment with the highest AP equal to 0.860 compared to 0.829. Considering both AP values are above 0.8, both experiments performed relatively well for the call detection system.

5.2 Experiment 2: MFCC Features

For the mel-frequency cepstral coefficients (MFCC) model, three main experiments were considered. The first experiment used the MFCC output with no added derivatives for the feature. The second experiment used the original 13 feature coefficients from experiment 1 and added 13 more coefficients in the form of the derivatives of the original 13. The third and final experiment considered the 26 coefficients used in the second experiment and added another 13 in the form of the second derivatives of the original 13 MFCC output coefficients.

For each experiment, two sub-experiments were performed. The first was performed using 5 templates and the second using 10 templates.

Table 5.4: Calibration results for the MFCC model: f1-scores and thresholds per call type, average f1-score and selected frame length (in seconds) for each feature set (no derivatives, first derivatives and first and second derivatives) with 5 and 10 templates per call type.

	Number of Templates	Single Tones		Multi Tones		Burst Tonals		Average F1	Frame Length
		F_1	Threshold	F_1	Threshold	F_1	Threshold		
MFCC	5	0.718	16.394	0.675	22.688	0.500	19.483	0.631	0.384
	10	0.784	17.507	0.402	19.553	0.472	21.288	0.553	0.256
MFCC deriv1	5	0.683	15.610	0.659	23.440	0.500	20.337	0.614	0.384
	10	0.785	18.815	0.394	22.548	0.425	20.569	0.535	0.256
MFCC deriv2	5	0.758	18.199	0.659	24.977	0.518	21.595	0.600	0.384
	10	0.806	19.642	0.408	22.356	0.430	22.836	0.548	0.256

As seen in Table 5.4, all three experiments for the MFCC model chose the same frame length for the 5- and 10-template sub-experiments. These values were 0.384 and 0.256 seconds for the 5- and 10-template sub-experiments respectively. This suggests a possible link between the number of templates per call type and the chosen frame length, as the frame length changed with the number of templates but not with the number of coefficients used.

For all three experiments the 5-template sub-experiment achieved a higher average f1-score over all three thresholds. In terms of average f1-score over all three thresholds, the MFCC experiment without derivatives performed the best. The highest average f1-score was for the 5-template sub-experiment with a score of 0.631. The second highest average f1-score was for the experiment where the first derivatives were added, with a score of 0.614.

The lowest f1-score was for the 10-template sub-experiment for the experiment where the first derivatives were added, with a score of 0.535.

In terms of the 10-template sub-experiments, the experiment using only the first 13 MFCC output coefficients performed the best in terms of average f1-score with a score of 0.553. The second best f1-score across the three 10-template sub-experiments was for the third experiment using all 39 coefficients. This score was 0.548.

The increase in diversification of templates generally decreased the average f1-scores during threshold selection.

For all three experiments and their respective sub-experiments, the single-tone call type performed the best in terms of f1-score for its chosen threshold. The highest f1-score was 0.806 for the 10-template sub-experiment of the third experiment containing all 39 coefficients. This may be linked to the simplicity of the call type, as it only contains a single frequency component, unlike the other two call types. It is also important to note

that the jump between f1-scores for the single-tone call type and the other two call types was rather large, further supporting this argument.

Table 5.5: Vote fraction selection results for the MFCC model: precision, recall and f1-score at the chosen vote fraction for each feature set with 5 and 10 templates per call type.

	Number of Templates	Precision	Recall	F_1	Vote Fraction
MFCC	5	1.000	0.900	0.947	0.6
	10	0.935	0.967	0.951	0.5
MFCC deriv1	5	0.935	0.967	0.951	0.4
	10	0.964	0.900	0.931	0.6
MFCC deriv2	5	0.906	0.967	0.935	0.4
	10	1.000	0.900	0.947	0.6

In terms of the f1-score for the choice of vote fraction, as seen in Table 5.5, the highest score of 0.951 was shared by the 10-template sub-experiment for the MFCC experiment without derivatives and the 5-template sub-experiment for the MFCC experiment with the first derivatives added.

The second highest f1-score of 0.947 was shared by the 5-template sub-experiment for the MFCC experiment without derivatives and the 10-template sub-experiment of the MFCC experiment with two derivatives.

Considering that the MFCC experiment with no derivatives held both the first and second highest f1-scores, this model was seen to have performed best in terms of overall call detection at the chosen hyperparameter settings for the calibration data. This performance can only be considered in terms of the hyperparameter choices and not in terms of the overall performance of the model, due to the use of the same calibration templates for all hyperparameter selection choices.

The vote fractions chosen for the MFCC experiments with the first and second derivatives stayed constant in terms of number of templates. The 5-template sub-experiments chose a vote fraction of 0.4 (2 out of 5 detections) and the 10-template sub-experiments chose 0.6 (6 out of 10 detections). The MFCC experiment without derivatives had different vote fractions, with the 5-template sub-experiment choosing 0.6 (3 out of 5 detections) and the 10-template sub-experiment choosing 0.5 (5 out of 10 detections).

Table 5.6: Final evaluation results for the MFCC model: precision, recall, f1-score and average precision (AP) for each feature set with 5 and 10 templates per call type.

	Number of Templates	Precision	Recall	F_1	Average Precision
MFCC	5	0.582	0.714	0.641	0.642
	10	0.535	0.817	0.647	0.684
MFCC deriv1	5	0.495	0.799	0.611	0.632
	10	0.468	0.809	0.593	0.602
MFCC deriv2	5	0.354	0.880	0.505	0.614
	10	0.505	0.770	0.610	0.628

For the MFCC experiment with no derivatives, the 5-template sub-experiment performed the best for precision. The 10-template sub-experiment performed best in terms of recall and f1-score. The highest values for precision, recall and f1-score respectively were 0.582, 0.817 and 0.647.

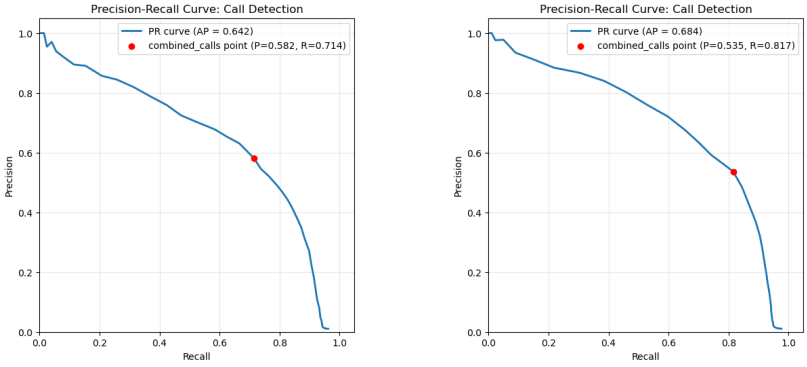
For the MFCC experiment with the first derivatives added, the 5-template sub-experiment performed best in precision and f1-score, while the 10-template sub-experiment performed best for recall. The highest values for precision, recall and f1-score respectively were 0.495, 0.809 and 0.611.

For the MFCC experiment with the second derivatives added, the 5-template sub-experiment performed best for recall. The 10-template sub-experiment performed best for precision and f1-score. The highest values for precision, recall and f1-score respectively were 0.505, 0.880 and 0.610.

Overall, in terms of f1-score, the 10-template sub-experiment for the MFCC experiment with no derivatives performed the best. By looking at the values in Table 5.6, it can be seen that the highest overall hyperparameter selection performance is characterised by balancing precision and recall and not by getting the highest value in either category.

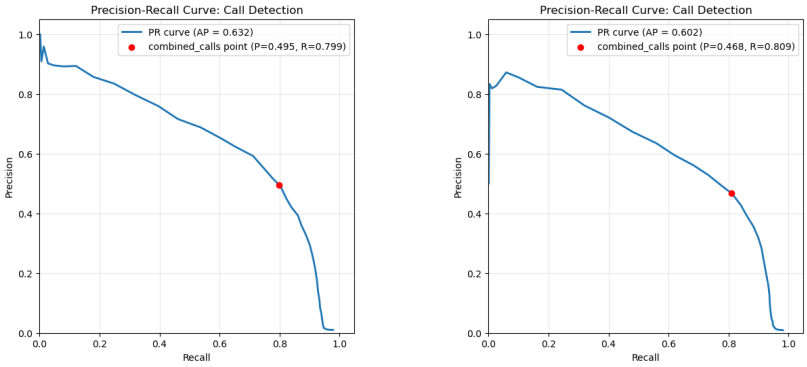
The precision for all experiments was consistently much lower than recall. This shows that the model was more lenient than what it was tuned for.

In terms of average precision (AP) and the precision-recall (PR) curve, the MFCC experiment with no derivatives outperformed all other MFCC experiments with a maximum AP of 0.684 for the 10-template sub-experiment. This can be seen by looking at both Table 5.6 and Figures 5.2, 5.3 and 5.4. Considering the AP was below 0.7, this was not considered a good performance. In terms of the hyperparameters selected for the model, all experiments did relatively well, as the PR points were in the best position given the model capabilities.



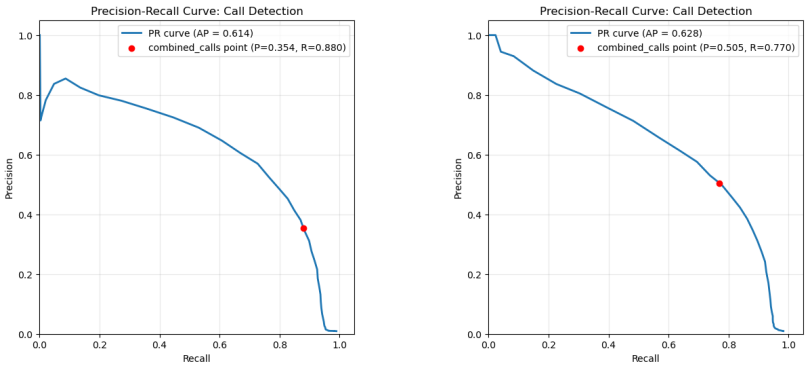
(a) 5 templates **(b)** 10 templates

Figure 5.2: The PR curves for MFCC



(a) 5 templates **(b)** 10 templates

Figure 5.3: The PR curves for MFCC with the first derivative added



(a) 5 templates **(b)** 10 templates

Figure 5.4: The PR curves for MFCC with the first and second derivatives added

5.3 Experiment 3: LPCC Features

For the linear predictive coding coefficients (LPCC) model, two experiments were conducted. The first considered the use of 5 templates and the second considered the use of 10 templates for the per call type database for hyperparameter selection and evaluation.

Table 5.7: Calibration results for the LPCC model: f1-scores and thresholds per call type, average f1-score and selected frame length (in seconds) for 5 and 10 templates per call type.

Number of Templates	Single Tones		Multi Tones		Burst Tonals		Average F1	Frame Length
	F_1	Threshold	F_1	Threshold	F_1	Threshold		
5	0.863	0.344	0.744	0.526	0.632	0.449	0.746	0.256
10	0.882	0.336	0.605	0.364	0.686	0.427	0.724	0.256

Both the 5- and 10-template experiments for the LPCC model chose 0.256 seconds as the frame length. This can be seen in Table 5.7. This shows that template diversification did not have a large effect on the selected frame length.

The 5-template experiment had the highest average f1-score of 0.746. However, the two average f1-scores were very similar, therefore template diversification had little effect on threshold selection.

The f1-scores for the single-tone call type for both experiments were much higher than for burst-tonals and multi-tones. This is possibly due to the simplicity of the single-tone structure.

Table 5.8: Vote fraction selection results for the LPCC model: precision, recall and f1-score at the chosen vote fraction for 5 and 10 templates per call type.

Number of Templates	Precision	Recall	F1	Vote Fraction
5	0.968	1.000	0.984	0.4
10	0.968	1.000	0.984	0.5

While the 5- and 10-template experiments for LPCC each chose a different vote fraction, in terms of precision, recall and f1-score they performed equally well. The chosen vote fraction for each was 0.4 and 0.5 for the 5- and 10-template experiments respectively. This means 2 of 5 templates and 5 of 10 templates had to agree for a detection to occur.

The f1-score for both experiments was 0.984. This can be seen in Table 5.8. This f1-score carries no weight in terms of the actual performance of the model when tested against evaluation data.

Despite the difference in vote fractions, the increase in diversification had no effect on vote fraction selection performance.

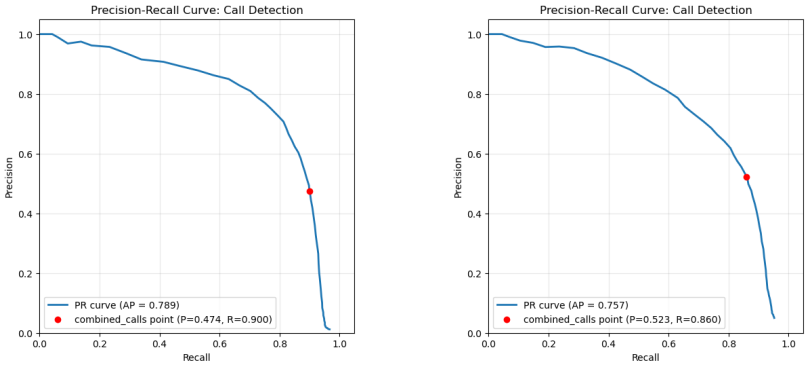
Table 5.9: Final evaluation results for the LPCC model: precision, recall, f1-score and average precision (AP) for 5 and 10 templates per call type.

Number of Templates	Precision	Recall	F1	Average Precision
5	0.474	0.900	0.621	0.789
10	0.523	0.860	0.650	0.757

As seen in Table 5.9, the 5-template experiment performed better in terms of recall. The 10-template experiment performed best in precision and f1-score. The highest precision, recall and f1-score were 0.523, 0.900 and 0.650 respectively.

This means that the 10-template experiment performed better in terms of chosen thresholds. This can also be seen in Figures 5.5a and 5.5b, where the marked precision and recall point is better matched to the precision-recall (PR) curve for the 10-template experiment.

The precision for both experiments is much lower than the recall. This shows the model is lenient in call detection.



(a) 5 templates **(b)** 10 templates

Figure 5.5: The PR curves for LPCC

By considering Table 5.9 and Figures 5.5a and 5.5b, it can be seen that in terms of average precision (AP) and PR curve shape, the 5-template experiment performed better with an AP value of 0.789.

Even though the 10-template experiment gave a better PR point due to chosen thresholds, the 5-template experiment for the LPCC model performed best overall in terms of Bryde's whale call detections due to the PR curve shape and AP value. However, considering both AP values were above 0.7, the model is seen to perform relatively well.

5.4 General Observations

The lack of data and human error meant that a maximum of 10 templates per call type could be used for call type calibration templates. It also caused a bias in the choice of templates to train on, as selecting suitably representative templates per call type became difficult. The selection of background calibration templates meant that the class imbalance between calls and background noise differed between hyperparameter selection and evaluation. The selection of templates, and not a continuous audio file, for hyperparameter selection also meant that subsequence DTW did not necessarily act under the same conditions as in evaluation, thus creating another difference between the two datasets.

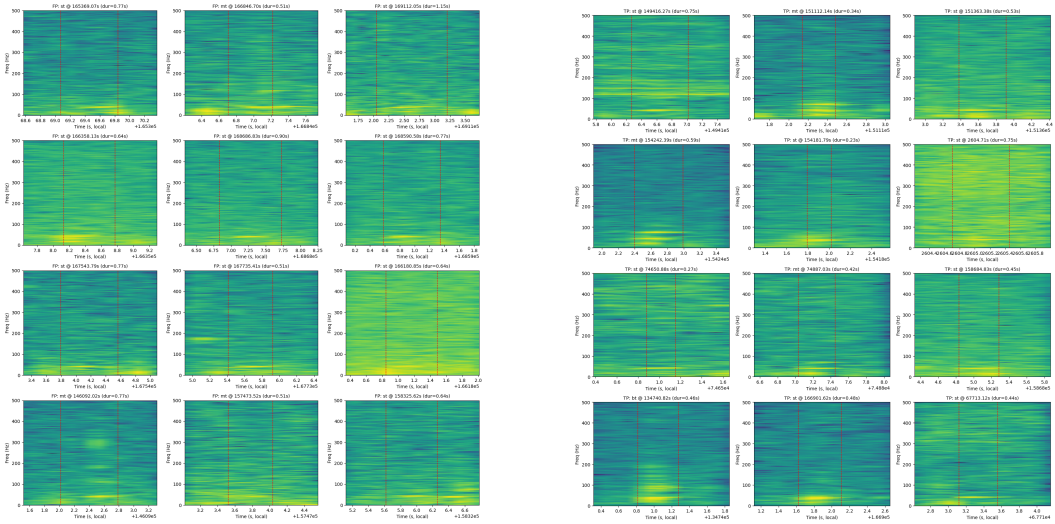
The varying performance of the calibrated thresholds is consistent with the low availability of well-labelled data to use for the hyperparameter selection stage of this project, as well as human error, which was introduced in the manual selection of calibration templates. This varying performance can be seen in Tables 5.1, 5.4 and 5.7.

Table 5.10: Summary of the highest f1-score and AP achieved by each feature extraction method, with the corresponding number of templates per call type.

Feature extraction method	F1		AP	
	Highest value	Number of Templates	Highest value	Number of Templates
STFT	0.792	5	0.860	10
MFCC	0.647	10	0.684	10
LPCC	0.650	10	0.789	5

By looking at Table 5.10, it can be seen that the STFT model outperforms the other models in terms of both f1-score and AP. This means that, both in terms of the model's ability to select hyperparameters and in terms of the efficiency of the detection pipeline, the STFT model outperformed the other two models. The LPCC model comes second, but in terms of precision it is not much better than the MFCC model.

When looking at printed spectrograms of false positives (FP) detected by the 10-template STFT model, it was found that for both experiments for the best performing model, many of these FP calls were actually unlabelled calls. This means that the actual precision of the model may be higher than the reported amounts. This may be due to the human error introduced in call labelling, since the known labelled calls were found by previous projects and were therefore not necessarily a complete list. Examples of FP calls compared to similar true positive (TP) calls can be seen in Figure 5.6.

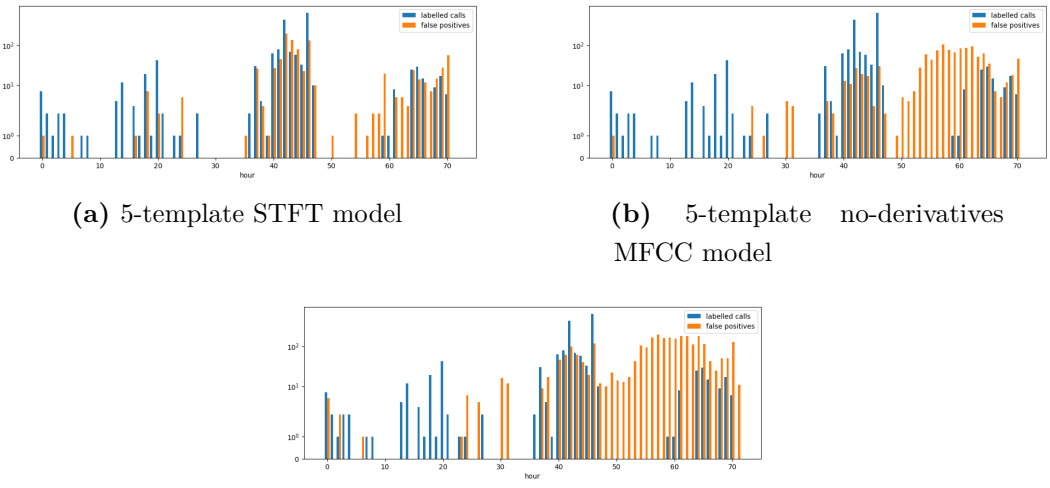


(a) False Positive samples

(b) True Positive samples

Figure 5.6: Examples of random samples of FP and TP for which some FP could be unlabelled calls for the 10-template experiment for the STFT model.

Another observation across all three detection models was that most of the FP calls were found within the second half of the 72-hour audio file used for the evaluation stage of this project. This implies a possible link between model performance and the signal-to-noise ratio (SNR) of the section of audio considered. The increase in FPs also seems to correlate with the increase in labelled calls. These observations can be seen in Figure 5.7.



(a) 5-template STFT model

(b) 5-template no-derivatives MFCC model

(c) 5-template LPCC model

Figure 5.7: This shows for each model where FP calls were detected in relation to labelled calls over the entire 72-hour audio file.

Considering the good performance of the model as well as the unlabelled TP calls, the STFT feature extraction is the best method to use for call detection with the DTW

algorithm, and the algorithm is a valid choice of detection method for Bryde's whales within the False Bay region.

5.5 Chapter Summary

In this chapter the results for all conducted experiments were discussed (STFT experiments, MFCC experiments and LPCC experiments). In the next chapter, conclusions will be drawn based on observations made from the results.

Chapter 6

Conclusion

6.1 Summary

The use of dynamic time warping (DTW) as a Bryde's whale call detection system was investigated in this project. Three feature extraction methods were tested: short-time Fourier transform (STFT), mel-frequency cepstral coefficients (MFCC) and linear predictive coding coefficients (LPCC).

6.2 Objectives

The first objective of this project was to implement a DTW-based detection system using three feature extraction methods: the short-time Fourier transform (STFT), mel-frequency cepstral coefficients (MFCCs) and linear predictive coding coefficients (LPCC). This was designed and implemented as explained in Chapter 4.

The second objective was to tune the feature frame length, call type thresholds and vote fraction for each method based on f1-score. The results for this can be seen in Tables 5.1, 5.2, 5.4, 5.5, 5.7 and 5.8.

The third objective was to evaluate the detection performance of each configuration using precision, recall, f1-score and average precision (AP). This can be seen in Tables 5.3, 5.6 and 5.9.

6.3 Limitations

Limitations in this project included the scarcity of well-labelled data and the human error involved in the labelling, due to a range of human labellers being used for different sections. This meant that it was necessary to select call type and background calibration templates, which caused more human error within the template selection stage. The necessity of

calibration template selection led to an imbalance between calls and background noise in hyperparameter tuning and evaluation. The labelling based on previous projects also led to many false positives (FPs) actually being calls that were never labelled.

6.4 Conclusion

The STFT model performed the best out of the three considered models. The maximum f1-score across all STFT experiments was 0.792 and the maximum AP was 0.860. The LPCC model performed second best with best scores of 0.650 and 0.789 for f1-score and AP respectively. The MFCC model performed the worst with a highest f1-score and AP value of 0.647 and 0.684 respectively.

For the MFCC experiments the addition of derivatives did not improve the model. For all three models an increase in template diversification (5 vs 10 templates per call type) had little effect on hyperparameter selection. It was found that the single-tone thresholds chosen for all experiments were generally better than the other thresholds, with a possible link to the simplicity of their frequency structure in comparison to the other two call types. Multi-tone call thresholds always had the lowest f1-scores, probably due to human error caused by different definitions decided by each labeller.

Precision was consistently lower than recall, showing that all models were more lenient in the evaluation stage than intended during the hyperparameter selection stage. Inspection of FP calls for the STFT model found that many of the FP calls were actually unlabelled calls. This shows that the limitations may have caused a skewed precision result in the evaluation analysis. The presence of FP calls closer to already labelled calls and in the second half of the 72-hour file pointed to possible connections between SNR and call density and FP detections.

Overall the STFT model was the most suitable method by a sizeable margin in both f1-score and AP values for detection of Bryde's whale calls in the False Bay region.

6.5 Future Work

More consistent labelling and more in-depth labelling of all audio files in future would be beneficial. A model could be tuned on full audio files instead of selected templates, and many of the limitations would be removed. The consideration of different methods to remove background noise would also be beneficial.

Once the limitations of the current detection model have been addressed, the next step would be to extend the system to classify the call type of each detection in continuous audio files.

References

[1] G. Tonitz, C. M. Geldenhuys, and T. R. Niesler, "A perch out of water: Few-shot detection of bryde's whale vocalisations in false bay," 2026, https://arxiv.org/abs/2610.29572.

[2] K. van Waerebeek, A. N. Baker, F. Félix, J. Gedamke, M. Iñiguez, G. P. Sanino, E. Secchi, D. Sutaria, A. van Helden, and Y. Wang, "Vessel collisions with small cetaceans worldwide and with large whales in the southern hemisphere; building a standardized database," 2006, https://www.vliz.be/imisdocs/publications/243193.pdf.

[3] A. Faustmann, F. W. Shabangu, and J. D. J. J. Versfeld, "Potential vocalizations of bryde's whales in false bay, south africa," 2024, https://repository.up.ac.za/server/api/core/bitstreams/57ba6a31-7aca-4acb-b9b6-60884695c964/content.

[4] S. Betz, K. Bohnsack, A. R. Callahan, L. E. Campbell, S. E. Green, and K. M. Labrum, "Reducing the risk of vessel strikes to endangered whales in the santa barbara channel," 2011, https://nmssanctuaries.blob.core.windows.net/sanctuaries-prod/media/archive/protect/shipstrike/pdfs/bren.pdf.

[5] V. B. Deecke, J. K. B. Ford, and P. Spong, "Call diversity in the north pacific killer whale populations: implications for dialect evolution and population history," *Animal Behaviour*, 2000, https://www.sciencedirect.com/science/article/abs/pii/S0003347211005537.

[6] M. A. McDonald, S. L. Mesnick, and J. A. Hildebrand, "The geographic variation of blue whale song," 2003, https://www.cetus.ucsd.edu/docs/reports/McDonaldIWC-2003.pdf.

[7] J. C. Brown, A. Hodgins-Davis, and P. J. O. Miller, "Classification of vocalizations of killer whales using dynamic time warping," 2006, https://pubs.aip.org/asa/jasa/article/119/3/EL34/850621.

[8] J. Herreilers, C. Jacobs, and T. Niesler, "Low-resource keyword spotting using contrastively trained transformer acoustic word embeddings," 2025, https://arxiv.org/pdf/2506.17690.

[9] E. M. Oleson, J. Barlow, J. Gordon, S. Rankin, and J. A. Hildebrand, "Low frequency calls of bryde's whales," 2003, https://digitalcommons.unl.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1297&context=usdeptcommercepub.

[10] M. Müller and F. Zalkow, "Fundamentals of music processing: Dynamic time warping (dtw)," 2015, https://www.audiolabs-erlangen.de/resources/MIR/FMP/C3/C3S2_DTWbasic.html.

[11] H. Kamper, "Dynamic time warping (dtw)," 2021, https://www.youtube.com/playlist?list=PLmZlBIcArwhMJoGk5zpiRlkaHUqy5dLzL.

[12] R. Schwaber-Cohen, "Vector similarity explained," 2023, https://www.pinecone.io/learn/vector-similarity/.

[13] T. S. community, "scipy.spatian.distnace," 2008, https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.cdist.html.

[14] E. Alizadeh, "An illustrative explanation to dynamic time warping," 2020, https://towardsdatascience.com/an-illustrative-introduction-to-dynamic-time-warping-36aa98513b98/.

[15] M. Müller and F. Zalkow, "Fundamentals of music processing: Subsequence dtw," 2015, https://www.audiolabs-erlangen.de/resources/MIR/FMP/C7/C7S2_SubsequenceDTW.html.

[16] D. J. J. V. Atinde M. Usman, Olayinka O. Ocundile, "Review of automatic detection and classification techniques for cetacean vocalization," 2020, https://scholar.sun.ac.za/server/api/core/bitstreams/6e4a861d-74eb-4d2d-b81b-353743cc06b8/content.

[17] H. Kamper, "Short-time fourier transforms (stft)," 2026, https://www.youtube.com/watch?v=tQeDi2qEW-s&list=PLmZlBIcArwhMrUZ_njMmlO_T8gAWHihNS&index=26.

[18] ——, "Introduction to speech features," 2021, https://www.youtube.com/playlist?list=PLmZlBIcArwhN8nFJ8VL1jLM2Qe7YCcmAb.

[19] T. Singh, "Mfcc's made easy," 2019, https://medium.com/@tanveer9812/mfccs-made-easy-7ef383006040.

[20] DataMListic, "Spectral features - deltas and delta-deltas explained," 2023, https://www.youtube.com/watch?v=zxEnuPolylY.

[21] ——, "Mel frequency cepstral coefficients (mfcc) explained," 2023, https://www.youtube.com/watch?v=SJo7vPgRlBQ.

[22] S. A. Majeed, H. Husain, S. A. Samad, and T. F. Idbeaa, "Mel frequency cepstral coefficients (mfcc) feature extraction enhancement in the application of speech recognition: A comparison study," 2015, https://www.jatit.org/volumes/Vol79No1/5Vol79No1.pdf.

[23] L. Rabiner and B.-H. J. Juang, *Fundamentals of Speech Recognition*. Prentice Hall, 1993.

[24] J. C. Brown and P. J. O. Miller, "Automatic classification of killer whale vocalizations using dynamic time warping," 2007, https://pubs.aip.org/asa/jasa/article-abstract/122/2/1201/699104/Automatic-classification-of-killer-whale?redirectedFrom=fulltext&casa_token=OzamuY0etVkAAAAA:VNGCuA1SmTnQEYgVRN2KuKcJvbDF79bLT2VkmggZSIUc2qnhYfNt_Yo7BleyPL4ffIKE3CTopQ.

[25] O. O. Ogundile and D. J. J. Versfeld, "Analysis of template-based detection algorithms for inshore bryde's whale short pulse calls," 2020, https://ieeexplore.ieee.org/document/8957435.

[26] X. Mouy, M. Bahoura, and Y. Simard, "Automatic recognition of fin and blue whale calls for real-time monitoring in the st. lawrence," 2009, https://pubs.aip.org/asa/jasa/article-abstract/126/6/2918/627068/Automatic-recognition-of-fin-and-blue-whale-calls?redirectedFrom=fulltext&casa_token=_uPiaa9QXCsAAAAA:Ge3i-ujMkyY8PsWGIQ4iGgW0FRxsPD0VtXtwuCMONoJvGUJFQAuBTQu8Iq7TWFSif-LMp-Eos

[27] O. O. Ogundile, A. Usman, O. Babalola, and D. Versfeld, "A hidden markov model with selective time domain feature extraction to detect inshore bryde's whale short pulse calls," 2020, https://www.sciencedirect.com/science/article/pii/S1574954120300376?casa_token=6Nq5H72ZrPQAAAAA:X0axwemDJS8TgltJ62x5518DbviKiMtXRjb7HX_TJkOnpH1IpbmheAeQcZnjfbrx6TNF-oP_Z7o.

[28] Y. Permanasari, E. H. Harahap, and E. P. Ali, "Speech recognition using dynamic time warping (dtw)," 2019, https://iopscience.iop.org/article/10.1088/1742-6596/1366/1/012091/meta.

[29] A. Faustmann, "Potential vocalizations of bryde's whales in false bay, south africa," 2024, https://scholardata.sun.ac.za/articles/dataset/An_Analysis_of_Potential_Low_Frequency_Vocalizations_of_Southern_African_Bryde_s_Whales_/25108016?file=44351747.

[30] B. Meddins, "The nyquist frequency," 2000, https://www.sciencedirect.com/topics/computer-science/nyquist-frequency.

[31] A. M. Learning, "Windowing functions explained," 2026, https://apxml.com/courses/introduction-to-speech-recognition/chapter-2-processing-audio-signals/windowing-functions-explained.

[32] librosa development team, "librosa.feature.mfcc," 2026, https://librosa.org/doc/main/api/generated/librosa.feature.mfcc.html.

[33] T. S. community, "dct," 2008, https://docs.scipy.org/doc/scipy/reference/generated/scipy.fft.dct.html.

[34] M. A. Roch, A. Širović, and S. Baumann-Pickering, "Detection, classification, and localization of cetaceans by groups at the scripps institution of oceanography and san diego state university," 2013, https://www.cetus.ucsd.edu/docs/publications/RochDCL2013.pdf.

[35] librosa development team, "librosa.feature.delta," 2023, https://librosa.org/doc/dev/api/generated/librosa.feature.delta.html.

[36] B. McFee, C. Raffel, D. Liang, D. P. Ellis, M. McVicar, E. Battenberg, and O. Nieto, "librosa: Audio and music signal analysis in python," 2015, https://proceedings.scipy.org/articles/Majora-7b98e3ed-003.pdf.

[37] librosa development team, "librosa.util.frame," 2025, https://librosa.org/doc/0.11.0/generated/librosa.util.frame.html.

[38] ——, "librosa.lpc," 2025, https://librosa.org/doc/0.11.0/generated/librosa.lpc.html.

[39] ——, "librosa.sequence.dtw," 2026, https://librosa.org/doc/0.11.0/generated/librosa.sequence.dtw.html.

[40] P. Hees, "The fast fourier transform, how and why it works," 2026, https://research.ibm.com/blog/the-fast-fourier-transform-how-and-why-it-works.

[41] B. Bechtold, "soundfile 0.14.0," 2026, https://pypi.org/project/soundfile/.

[42] T. S. community, "resample_poly," 2008, http://docs.scipy.org/doc/scipy/reference/generated/scipy.signal.resample_poly.html.

[43] ——, "find_peaks," 2008, https://docs.scipy.org/doc/scipy/reference/generated/scipy.signal.find_peaks.html.

[44] N. Developers, "numpy.searchsorted," 2026, https://numpy.org/doc/stable/reference/generated/numpy.searchsorted.html.

[45] T. Fawcett, "An introduction to roc analysis," 2005, https://www.sciencedirect.com/science/article/pii/S016786550500303X.

[46] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets," 2015, https://www.researchgate.net/publication/273155496_The_Precision-Recall_Plot_Is_More_Informative_than_the_ROC_Plot_When_Evaluating_Binary_Classifiers_on_Imbalanced_Datasets.

[47] R. Padilla, S. L. Netto, and E. A. B. da Silva, "A survey on performance metrics forobject-detection algorithms," 2020, https://www.researchgate.net/publication/343194514_A_Survey_on_Performance_Metrics_for_Object-Detection_Algorithms.

Appendix A

Project Planning Schedule

Table A.1: Project Schedule

Schedule	Tasks
20/07/2026 - 02/08/2026	Research DTW, STFT, MFCC and LPC as well as related work
03/08/2026 - 16/08/2026	Download data and experiment with feature extraction and basic DTW
17/08/2026 - 23/08/2026	Implement DTW functionality for STFT feature extraction
24/08/2026 - 06/09/2026	Implement hyperparameter selection stage for STFT
07/09/2026 - 27/09/2026	Implement evaluation stage for STFT
28/09/2026 - 11/10/2026	Implement MFCC and LPC feature extraction methods on current model
12/10/2026 - 25/10/2026	Ensure all three models work and make any changes necessary
26/10/2026 - 31/10/2026	Finish Report and submit

Appendix B

Graduate Attributes

B.1 Complex Engineering Problem

A systematic, theory-based method was used to design the Bryde's whales call detection system. The systematic approach can be seen in the report structure. Engineering specialist knowledge can be seen in the theory and mathematics behind all engineering concepts needed for this project. This was researched and described in Chapter 2. After understanding the concepts a systematic approach was used for all design decisions. This included backing all design choices with sources or by testing as hyperparameters. This can be seen in Chapter 4.

As seen in Chapter 4 efficient use of resources was considered, thus, the 72-hour audio file which had to be processed was split into 1-hour segments for analysis to ensure efficient use of device memory.

As seen in Chapter 3 the data used for the project came from previous projects, therefore resources were re-used instead of capturing new data. The real world uses of a whale call detection system were considered. A system such as this could be used to protect a Bryde's whale colony from harm caused by shipping vessels. This was explained in Chapter 1.

Engineering technology used in this project included libraries such as Python, Librosa, SciPy and NumPy.

In Chapter 2 articles considering different approaches to a similar problem or using similar methodologies for different problems were considered.

As explained in Chapter 3 technical issues within this project included a scarcity in well labelled data which required creative thinking and planning for the hyperparameter selection stage. These issues required the solution of sub-problems (such as the data scarcity) in order to solve the larger problem.

B.2 GA 1: Problem Solving

As explained in Chapter 1 and 3 the problem of Bryde's whale call detection in a continuous audio file with a scarcity of labelled data was identified. Literature relating to the problem was considered in the form of other implementations of similar solutions. The theory behind solution decisions was studied and explained in detail. These topics include DTW, subsequence DTW, STFT, MFCC and LPC as well as precision-recall (PR) analysis. The literature and theoretical explanations were explained in Chapter 2.

Chapter 2 also shows understanding of the whale call structure and its effect on decision made for the detection pipeline.

Chapter 1 showed that sustainability impact of the project was considered in the form of the uses of such a detection system to prevent the harm of Bryde's whales in False Bay.

The analysis of the detection system was done using precision, recall, f1-score and average precision (AP). These are all topics researched and explained in Chapter 4. The use of these analysis methods to draw conclusions can be seen in Chapter 5 and 6.

B.3 GA 2: Application of Scientific and Engineering Knowledge

Mathematics, natural sciences and engineering sciences are applied in formal analysis and modelling of the engineering problem. This can be seen in the theoretical and mathematical explanations given in Chapter 2 as well as in the decisions and implementation explained in Chapter 4.

B.4 GA 3: Engineering Design

A Bryde's whale call detection system was designed with a scarcity of well labelled data. All design choices were made based on theory studied. The choices for which values to test for the tuned hyperparameters was also based on research. All choices were therefore made with consideration and required creative thinking. This can be seen in Chapter 4

B.5 GA 4: Investigations, experiments and data analysis

In Chapter 5 and 6 it is explained how research based analysis methods were used to draw conclusions on the effectiveness on the DTW detection system for different feature

extraction methods. The methods used for evaluation (precision, recall, f1-score and AP) were researched and explained in Chapter 4.

B.6 GA 5: Use of Engineering Tools

In Chapter 2 techniques used for this project were explained. In Chapter 4 the implementation of these techniques using engineering tool such as Python and its libraries: Librosa, SciPy and NumPy.

B.7 GA 6: Professional and Technical Communication

All research, design, implementation and analysis of the project is explained within this academic report. This report includes explanations as well as tables, figures and software flow diagrams where appropriate. This can be seen in all chapters of this report.

B.8 GA 8: Individual Work

This project was undertaken as an individual assignment with Professor D.J.J. Versfeld as a supervisor. Planning of project timeline as shown Appendix A was conducted as individual work. All work conducted during the project was analysed and possible improvements were suggested in Chapter 6. This shows the ability for self-reflection.

B.9 GA 9: Independent Learning Ability

Self-learning was conducted to help define the exact problem considered as well as determine a solution. This knowledge gained can be seen in Chapter 2 and 4.

B.10 Sustainable Development

As explained in Chapter 3 the dataset was re-used from a previous project, therefore time and resources was not put into capturing new whale call recordings. This is in line with sustainable development.

As seen in Chapter 1 the use of such a detection system in a real world application of preventing harm to Bryde's whales in False Bay was considered and discussed. This could help reduce the environmental impacts of human activities in whale territory.

B.11 Critical Thinking

This can be seen throughout the design process and results formulation as shown in Chapter 4, 5 and 6

Critical thinking was also necessary to combat the data scarcity explained in Chapter 3.

B.12 Emerging Technologies

An incorporation of artificial intelligence (AI) with a call detection system such as the one described in this project could be beneficial for helping identify when whales enter vessel heavy areas of their migration region. With an AI prediction and alert system could be implemented along a colonies migration region such that vessel traffic can be diverted before Bryde's whales enter the vessel area.

Appendix C

Use of Artificial Intelligence (AI) Tools

C.1 AI Work Declaration

AI tools used were the free versions of Claude and Gemini.

Used to help find some sources for research purposes.

Used for implementation of template selection Python script. This script involved using the file containing labelled calls and printing spectrograms of n labelled calls for specific labels linearly throughout the 72-hour audio file used for template selection.

Used for generating spectrograms, PR curve diagrams and false positive detections in relation to labelled calls for the full 72-hour file. Used for debugging throughout the coding process.

Used for spelling and grammar check after report was written. Used to determine any conflicting statements within the report.

C.2 Own Work Declaration

Read papers, website articles, textbook chapters and watched video explanations for research purposes. Only some of these sources were identified via AI, most were found with a simple Google search. After being referred to some sources via AI these sources were also externally found and studied.

Manually selected templates using the template selection Python script coded by AI. The spectrograms were manually moved through and templates were selected based on this human inspection.

Designed and implemented the hyperparameter selection stage of the project as well as the detection pipeline and calculation of evaluation metrics.

Wrote the project report and came to conclusions based on analysis metrics implemented during evaluation. Considered feedback on conflicting statements in the report and fixed

the relevant sections to remove these conflicts.